GLUING WEAK APPROXIMABILITY ALONG RECOLLEMENTS

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ABSTRACT. We study whether weak approximability glues along recollements of triangulated categories that are compactly generated by a single object. This is applied to derived categories of complexes with quasi-coherent cohomology and prescribed support for quasi-compact quasi-separated algebraic stacks with quasi-finite separated diagonal.

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1. INTRODUCTION

Approximable triangulated categories, and their weak variant, were introduced by Neeman [Nee21a, Nee21b, Nee26]. The theory has been used to settle conjectures of Bondal–Van den Bergh [Nee21b] and of Antieau–Gepner–Heller [Nee24]. It has also found applications to Fourier–Mukai theory for singular schemes [DLM25, GLMP25, GL26]. See [CNS25] for a survey of recent developments.

Roughly speaking, a triangulated category with small coproducts and a compact generator G is weakly approximable if every object in the aisle of a suitable t-structure can be suitably approximated from small coproducts of a uniformly bounded range of shifts of G. For example, if X is a quasi-compact quasi-separated scheme and $Z \subseteq X$ is a closed subset with quasi-compact complement, $D_{\mathrm{qc},Z}(X)$ is weakly approximable [Nee24, Theorem 3.2(iv)].

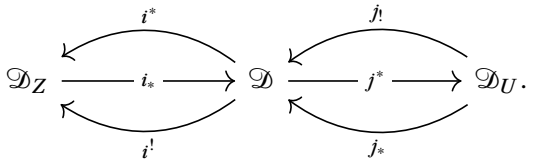
The main result is the following.

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Theorem 1.1. *Consider a recollement of singly compactly generated triangulated categories:*



Assume that $\mathcal{D}_Z$ and $\mathcal{D}_U$ are weakly approximable. Suppose G' is a compact generator of $\mathcal{D}$. Then $\mathcal{D}$ is weakly approximable if, and only if, $\mathrm{Hom}(G', G'[n]) = 0$ *for* $n \gg 0$.

The proof of [DLMP25, Theorem 1.2] does not rely on weak approximability of $D_{\mathrm{qc},Z}$ for algebraic stacks. As an application of Theorem 1.1, we establish weak approximability for quasi-compact quasi-separated algebraic stacks with quasi-finite separated diagonal.

Corollary 1.2. *Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack with quasi-finite separated diagonal. Suppose $S \subseteq |\mathcal{X}|$ is a closed subset with quasi-compact complement. Then $D_{\mathrm{qc},S}(\mathcal{X})$ is weakly approximable. In particular, $D_{\mathrm{qc}}(\mathcal{X})$ is weakly approximable. If $\mathcal{X}$ is Noetherian, then $D_{\mathrm{qc},S}(\mathcal{X})$ is coherent.*

Theorem 1.1 is the analog for weak approximability of [BNP23, Theorem 5.1] by Burke–Neeman–Pauwels. Loc. cit. shows that $\mathcal{D}$ is approximable if $\mathcal{D}_Z$ and $\mathcal{D}_U$ are approximable and $\mathrm{Hom}(G', G'[n]) = 0$ for $n \gg 0$. The 'only if' direction of Theorem 1.1 is formal (cf. [BNP23, Remark 3]). Thus, the work lies in the 'if' direction.

The proof in [BNP23] lifts approximations from $\mathcal{D}_Z$ to $\mathcal{D}$. This is done by induction on the number of cones needed to build them [BNP23, Lemmas 5.2 to 5.5]. At each step, the bound on the lift is worse than on its target. Hence, the constants grow with the number of cones. Approximability of $\mathcal{D}_Z$ bounds this number.

No such bound need exist if $\mathcal{D}_Z$ is only weakly approximable. Instead, we lift a single shift of a compact generator so that it lies in the same shifted aisle as its target (e.g. $\mathcal{D}^{\leq K}$). See Lemma 4.6. Consequently, the cone of a lift satisfies the same bound, which gives control on the desired objects. See Proposition 4.8.

We discuss Corollary 1.2. In general for schemes, $D_{\mathrm{qc},Z}$ need not be approximable [Nee24, Remark 8.1]. Hence, [BNP23, Theorem 5.1] does not apply, and the weak analog is necessary. The proof of Corollary 1.2 combines Neeman's result for schemes [Nee24, Theorem 3.2(iv)] with the induction principle for algebraic stacks [HR18, Theorem E], using finite flat descent and [Lan26, Corollary 3.6].

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AI disclosure. The authors used large language models (LLMs) to assist with proofreading, improving the exposition, checking arguments, and revising some proofs. While the main statements are due to the authors, parts of their proofs were developed in conversations with LLMs. Initially, this involved proposing to adapt the induction on the number of cones in [BNP23]. Subsequent conversations with LLMs on controlling the bounds led to lifting a single shift of a compact generator into the same shifted aisle as its target (see Lemma 4.6 and Proposition 4.8).

2. Preliminaries

2.1. **Triangulated categories**.

2.1.1. *Terminology.* We fix some terminology. See [Nee01, §1]. Let $\mathcal{T}$ be a triangulated category. Note that $\mathrm{Hom}(X,Y)$ is an abelian group. Given a morphism u, write $-u$ for its additive inverse. Consider a distinguished triangle

$$X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]. \tag{2.1}$$

Then TR2 asserts that

$$Y \xrightarrow{v} Z \xrightarrow{w} X[1] \xrightarrow{-u[1]} Y[1],$$

$$Z[-1] \xrightarrow{-w[-1]} X \xrightarrow{u} Y \xrightarrow{v} Z$$

are also distinguished. Passing from (2.1) to the first (resp. second) of these is called **rotating** (resp. **rotating backwards**). Observe that negating two of the three morphisms of a distinguished triangle gives a distinguished triangle. Indeed, $(1_X, -1_Y, 1_Z)$ is an isomorphism of triangles from (u,v,w) to $(-u,-v,w)$. For every $n \in \mathbb{Z}$, there exists a distinguished triangle

$$X[n] \xrightarrow{u[n]} Y[n] \xrightarrow{v[n]} Z[n] \xrightarrow{(-1)^n w[n]} X[n+1].$$

We call this **shifting** the given distinguished triangle by $[n]$. By TR1, every $f\colon X \to Y$ can be completed into a distinguished triangle

$$X \xrightarrow{f} Y \to C_f \to X[1].$$

We call C_f a **cone** of f. Rotating backwards gives

$$F_f \to X \xrightarrow{f} Y \to F_f[1].$$

We call $F_f := C_f[-1]$ a **cocone** of f. Both are unique up to (non-unique) isomorphism. By combining TR1 and TR2, it follows that every morphism can be realized as the first, second, or third of some distinguished triangle. Moreover, given a morphism of distinguished triangles, if two of the three vertical morphisms are isomorphisms, then so is the third. Indeed, this can be shown by rotating and applying [Nee01, Proposition 1.1.20].

2.1.2. *Generation.* Let $\mathcal{T}$ be a triangulated category admitting small coproducts and $\mathcal{S}, \mathcal{S}'$ full subcategories. Write $\mathcal{S} \star \mathcal{S}'$ for the full subcategory of objects X admitting a distinguished triangle $S \to X \to S' \to S[1]$ with $S \in \mathcal{S}$ and $S' \in \mathcal{S}'$. Write $\mathrm{Coprod}(\mathcal{S})$ for the smallest strictly full subcategory containing $\mathcal{S}$ that is closed under small coproducts and extensions, and $\mathrm{smd}(\mathcal{S})$ for the full subcategory of direct summands of objects of $\mathcal{S}$. For $H \in \mathcal{T}$ and integers $a \le b$ put

$$H[a,b] := \{H[-k] : a \le k \le b\}, \qquad \overline{\langle H \rangle}^{[a,b]} := \mathrm{smd}(\mathrm{Coprod}(H[a,b])).$$

The subcategory of compact objects is denoted by $\mathcal{T}^c$, consisting of those $c \in \mathcal{T}$ for which $\mathrm{Hom}(c,-)$ preserves coproducts. These form a triangulated subcategory of $\mathcal{T}$. The **localizing subcategory** generated by a collection $\mathcal{S} \subseteq \mathcal{T}$, denoted by $\overline{\langle \mathcal{S} \rangle}^{(-\infty,\infty)}$, is the smallest triangulated subcategory containing $\mathcal{S}$ and closed under small coproducts. If $\mathcal{T}^c$ is essentially small, then we say that $\mathcal{T}$ is **compactly generated** when $\mathcal{T} = \overline{\langle \mathcal{T}^c \rangle}^{(-\infty,\infty)}$. Equivalently, $\mathcal{T}$ is compactly generated if, for any $E \in \mathcal{T}$ satisfying $\mathrm{Hom}(P,E) = 0$ for all $P \in \mathcal{T}^c$, one has $E \cong 0$. An object $G \in \mathcal{T}^c$ is a **compact generator** of $\mathcal{T}$ if every $E \in \mathcal{T}$ with $\mathrm{Hom}(G[n],E) = 0$ for all $n \in \mathbb{Z}$ satisfies $E \cong 0$. An object of $\mathcal{T}^c$ is a **classical generator** of

$\mathcal{T}^c$ if the smallest thick subcategory of $\mathcal{T}^c$ containing it is $\mathcal{T}^c$. If $\mathcal{T}$ is compactly generated, then classical generators for $\mathcal{T}^c$ coincide with compact generators for $\mathcal{T}$ [Sta26, Tag 09SR]. We say that $\mathcal{T}$ is **singly compactly generated** if it admits a compact generator.

2.1.3. *t-structures.* Let $\mathcal{T}$ be a triangulated category with shift functor $[1]$. A pair of strictly full subcategories $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ of $\mathcal{T}$ is a ***t*-structure** if:

- $\mathrm{Hom}(A, B) = 0$ for all $A \in \mathcal{T}^{\leq 0}$ and $B \in \mathcal{T}^{\geq 0}[-1]$,
- $\mathcal{T}^{\leq 0}[1] \subseteq \mathcal{T}^{\leq 0}$ and $\mathcal{T}^{\geq 0}[-1] \subseteq \mathcal{T}^{\geq 0}$,
- for every $E \in \mathcal{T}$, there is a distinguished triangle
$$\tau^{\leq 0}E \to E \to \tau^{\geq 1}E \to (\tau^{\leq 0}E)[1]$$
with $\tau^{\leq 0}E \in \mathcal{T}^{\leq 0}$ and $\tau^{\geq 1}E \in \mathcal{T}^{\geq 0}[-1]$.

This notion was introduced in [BBDG18]. The above distinguished triangle is called the **truncation triangle** of E with respect to the t-structure. For any $n \in \mathbb{Z}$, $(\mathcal{T}^{\leq n}, \mathcal{T}^{\geq n})$ is also a t-structure on $\mathcal{T}$ where $\mathcal{T}^{\leq n} := \mathcal{T}^{\leq 0}[-n]$ and $\mathcal{T}^{\geq n} := \mathcal{T}^{\geq 0}[-n]$. Define the subcategories:

$$\mathcal{T}^{-} := \bigcup_{n=0}^{\infty} \mathcal{T}^{\leq n}, \quad \mathcal{T}^{+} := \bigcup_{n=0}^{\infty} \mathcal{T}^{\geq -n}, \quad \mathcal{T}^{b} := \mathcal{T}^{-} \cap \mathcal{T}^{+}.$$

Let $F\colon \mathcal{T} \to \mathcal{T}'$ be an exact functor between triangulated categories equipped with t-structures $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ and $((\mathcal{T}')^{\leq 0}, (\mathcal{T}')^{\geq 0})$. We say that F is **right** t**-exact** if $F(\mathcal{T}^{\leq 0}) \subseteq (\mathcal{T}')^{\leq 0}$, and **left** t**-exact** if $F(\mathcal{T}^{\geq 0}) \subseteq (\mathcal{T}')^{\geq 0}$. If both conditions hold, then F is t**-exact**.

A strictly full subcategory $\mathcal{A} \subseteq \mathcal{T}$ is an **aisle** if it is closed under positive shifts and extensions, and the inclusion $\mathcal{A} \to \mathcal{T}$ admits a right adjoint. Given a t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$, the subcategory $\mathcal{T}^{\leq 0}$ is an aisle. By [KV88, §1, Proposition], any aisle $\mathcal{A}$ determines a t-structure $(\mathcal{A}, \mathcal{A}^{\perp}[1])$, where

$$\mathcal{A}^{\perp} := \{T \in \mathcal{T} : \mathrm{Hom}(A, T) = 0 \text{ for all } A \in \mathcal{A}\}.$$

The subcategories $\mathcal{T}^{\leq 0}$ and $\mathcal{T}^{\geq 0}$ are called the **aisle** and **coaisle** of the t-structure, respectively. More generally, a **preaisle** is a strictly full subcategory closed under positive shifts and extensions. By [AJS03, Lemma 1.3 & Corollary 1.4], aisles are closed under direct summands and under small coproducts when these exist in $\mathcal{T}$.

Assume $\mathcal{T}$ admits small coproducts. A preaisle $\mathcal{A} \subseteq \mathcal{T}$ is called **cocomplete** if it is closed under all coproducts in $\mathcal{T}$. Given $\mathcal{S} \subseteq \mathcal{T}$, $\overline{\langle \mathcal{S} \rangle}^{(-\infty,0]}$ is defined to be the smallest cocomplete preaisle containing $\mathcal{S}$. By [Nee21c, Theorem 2.3], if $\mathcal{T}$ is well generated (e.g. compactly generated), $\overline{\langle \mathcal{S} \rangle}^{(-\infty,0]}$ is an aisle whenever $\mathcal{S}$ is essentially small. Furthermore, if $\mathcal{A} \subseteq \mathcal{T}^c$ is closed under positive shifts, then [CHNS24, Theorem 2.3.3 & Remark 2.3.4] (which generalizes [AJS03, Theorem A.1 & Proposition A.2]) says that $\mathrm{Coprod}(\mathcal{A})$ is the aisle $\overline{\langle \mathcal{A} \rangle}^{(-\infty,0]}$. An aisle $\mathcal{U}$ on $\mathcal{T}$ is **compactly generated** when there exists a collection of compact objects $\mathbb{Q} \subseteq \mathcal{T}^c$ satisfying $\overline{\langle \mathbb{Q} \rangle}^{(-\infty,0]} = \mathcal{U}$. In this case, we say $\mathcal{U}$ is compactly generated by $\mathbb{Q}$. We say a t-structure is **compactly generated** if its aisle is as such. For instance, if $G \in \mathcal{T}^c$, then

$$\mathrm{Coprod}(\{G[n] : n \geq 0\}) = \overline{\langle G \rangle}^{(-\infty,0]}.$$

Consider a pair of t-structures $(\mathcal{T}_1^{\leq 0}, \mathcal{T}_1^{\geq 0})$ and $(\mathcal{T}_2^{\leq 0}, \mathcal{T}_2^{\geq 0})$ on $\mathcal{T}$. We say they are **equivalent** if there exists $A \geq 0$ such that $\mathcal{T}_1^{\leq -A} \subseteq \mathcal{T}_2^{\leq 0} \subseteq \mathcal{T}_1^{\leq A}$. If $\mathcal{T}$ has a single compact

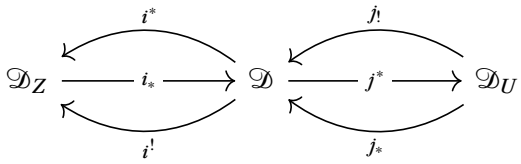
generator, the **preferred equivalence class** is the equivalence class of the t-structure compactly generated by any compact generator. By [Nee26, Remark 1.19], the preferred equivalence class does not depend on the choice of compact generator.

If $\mathcal{T}$ has small coproducts and is equipped with a t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$, let $\mathcal{T}_c^-$ be the collection of $E \in \mathcal{T}$ such that for every $n > 0$ there exists a distinguished triangle

$$D \to E \to F \to D[1]$$

with $D \in \mathcal{T}^c$ and $F \in \mathcal{T}^{\leq -n}$. Moreover, let $\mathcal{T}_c^b := \mathcal{T}^b \cap \mathcal{T}_c^-$. Furthermore, if $\mathcal{T}$ is weakly approximable and $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ is in the preferred equivalence class, then $\mathcal{T}^?$ and $\mathcal{T}_c^\#$ are independent of the t-structure representing the equivalence class for $? \in \{-, +, b\}$ and $\# \in \{-, b\}$. See [CHNS24, Remark 4.1.5].

2.1.4. *Recollements.* We briefly recall the notion of a recollement. See [BBDG18, §1.4] for details. A **recollement** is a diagram of triangulated categories and exact functors of the form



satisfying:

- $i^* \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$ (i.e. adjoint triples)
- $i_*, j_!, j_*$ are fully faithful
- $\ker(j^*)$ coincides with the strictly full subcategory on objects of the form $i_*(T)$ where $T \in \mathcal{D}_Z$.

In such a case, there are distinguished triangles

$$j_! j^* E \xrightarrow{\epsilon_E} E \xrightarrow{\eta_E} i_* i^* E \xrightarrow{\varphi_E} j_! j^* E[1],$$
$$i_* i^! E \to E \to j_* j^* E \to i_* i^! E[1]$$

which are functorial in $E \in \mathcal{D}$. In particular, the natural transformations between these functors are given by the (co)units of the relevant adjoint pairs. Since $j_!$, j^*, i_*, and i^* are left adjoints, they preserve coproducts. Additionally, if each triangulated category is compactly generated, i_* and j^* admitting right adjoints implies i^* and $j_!$ preserve compact objects [Nee96, Theorem 5.1]. Given $E \in \mathcal{D}_Z$ and $B \in \mathcal{D}_U$, $j^* i_* E \cong 0$, $i^! j_* B \cong 0$, and $i^* j_! B \cong 0$.

Assume $(\mathcal{D}_Z^{\leq 0}, \mathcal{D}_Z^{\geq 0})$ and $(\mathcal{D}_U^{\leq 0}, \mathcal{D}_U^{\geq 0})$ are t-structures respectively on $\mathcal{D}_Z$ and $\mathcal{D}_U$. It is possible to 'glue' these aisles to obtain a new t-structure on $\mathcal{D}$. Specifically, the glued t-structure is given by

$$\mathcal{D}^{\leq 0} := \left\{ E \in \mathcal{D} : i^* E \in \mathcal{D}_Z^{\leq 0}, j^* E \in \mathcal{D}_U^{\leq 0} \right\},$$
$$\mathcal{D}^{\geq 0} := \left\{ E \in \mathcal{D} : i^! E \in \mathcal{D}_Z^{\geq 0}, j^* E \in \mathcal{D}_U^{\geq 0} \right\}.$$

Moreover, a few general facts include: i^* and $j_!$ are right t-exact; i_* and j^* are t-exact; and $i^!$ and j_* are left t-exact. See [BBDG18, 1.4.10 & 1.3.17.(iii)] for details.

2.1.5. *Weak approximability.* We recall the definition of weak approximability for triangulated categories. See [Nee26, Definition 1.25].

Definition 2.1. A triangulated category $\mathcal{T}$ is **weakly approximable** if there exist a compact generator G, a t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ and an integer $A > 0$ such that:

(1) $G \in \mathcal{T}^{\leq A}$ and $\mathrm{Hom}(G, \mathcal{T}^{\leq -A}) = 0$;
(2) every $F \in \mathcal{T}^{\leq 0}$ admits a distinguished triangle $E \to F \to D \to E[1]$ with $E \in \overline{\langle G \rangle}^{[-A,A]}$ and $D \in \mathcal{T}^{\leq -1}$.

We call $(G, \mathcal{T}^{\leq 0}, A)$ **weak approximation data**.

In our convention, compactly generated triangulated categories a priori have small coproducts. In Definition 2.1, (1) is the condition $G[A] \in \mathcal{T}^{\leq 0}$, $\mathrm{Hom}(G[-A], \mathcal{T}^{\leq 0}) = 0$ of [BNP23, 2.9], since $\mathrm{Hom}(G[-A], \mathcal{T}^{\leq 0}) = \mathrm{Hom}(G, \mathcal{T}^{\leq -A})$.

A weakly approximable triangulated category $\mathcal{T}$ is **coherent** if, for any t-structure $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ in the preferred equivalence class, there exists an integer $N > 0$ such that every object $Y \in \mathcal{T}_c^-$ admits a distinguished triangle

$$X \to Y \to Z \to X[1]$$

with $X \in \mathcal{T}_c^- \cap \mathcal{T}^{\leq N}$ and $Z \in \mathcal{T}_c^b \cap \mathcal{T}^{\geq 0}$. See [CHNS24, Definition 10.1.1].

2.2. **Algebraic stacks.** We follow the conventions of [Sta26] for algebraic stacks. However, for the derived pullback and pushforward adjunction, we adopt the conventions of [HR17, §1]. Symbols such as X, Y, etc. denote schemes or algebraic spaces, whereas $\mathcal{X}$, $\mathcal{Y}$, etc. denote algebraic stacks. Fix a quasi-compact quasi-separated algebraic stack $\mathcal{X}$. Write $|\mathcal{X}|$ for its underlying topological space.

2.2.1. *Categories/functors.* $\mathrm{Mod}(\mathcal{X})$ denotes the Grothendieck abelian category of sheaves of $\mathcal{O}_{\mathcal{X}}$-modules on the lisse-étale site of $\mathcal{X}$. $\mathrm{Qcoh}(\mathcal{X})$ is the strictly full subcategory (i.e. full and closed under isomorphisms) of $\mathrm{Mod}(\mathcal{X})$ consisting of quasi-coherent sheaves. $D(\mathcal{X}) := D(\mathrm{Mod}(\mathcal{X}))$ is the derived category of $\mathrm{Mod}(\mathcal{X})$. $D_{\mathrm{qc}}(\mathcal{X})$ is the full subcategory of $D(\mathcal{X})$ consisting of complexes with quasi-coherent cohomology. $\mathrm{Perf}(\mathcal{X})$ denote the full subcategory of perfect complexes in $D_{\mathrm{qc}}(\mathcal{X})$. If $\mathcal{X}$ is Noetherian, then $D^?_{\mathrm{coh}}(\mathcal{X})$ for complexes with coherent cohomology where $? \in \{-, +, b\}$. For a morphism $f \colon \mathcal{Y} \to \mathcal{X}$ of quasi-compact quasi-separated algebraic stacks, we use the adjunction $\mathbf{L}f^* \dashv \mathbf{R}f_*$ on D_{qc} from [HR17, §1].

2.2.2. *Support.* For $M \in \mathrm{Qcoh}(\mathcal{X})$, let $\mathrm{supp}(M) := p\big(\mathrm{supp}(p^*M)\big) \subseteq |\mathcal{X}|$ where $p \colon U \to \mathcal{X}$ is any smooth surjective morphism from a scheme. This definition is independent of the choice of p. More generally, given $E \in D_{\mathrm{qc}}(\mathcal{X})$, let

$$\mathrm{supp}(E) := \bigcup_{j \in \mathbb{Z}} \mathrm{supp}\left(\mathscr{H}^j(E)\right) \subseteq |\mathcal{X}|.$$

This is called the **support** of E. Given an open immersion $j \colon \mathcal{U} \to \mathcal{X}$, $D_{\mathrm{qc},Z}(\mathcal{X}) := \ker(\mathbf{L}j^*)$ where $Z = |\mathcal{X}| \setminus |\mathcal{U}|$. It can be checked that

$$D_{\mathrm{qc},Z}(\mathcal{X}) = \{E \in D_{\mathrm{qc}}(\mathcal{X}) : \mathrm{supp}(E) \subseteq Z\}.$$

We use the analogous notation with coherent cohomology.

2.2.3. *Concentratedness.* A quasi-compact quasi-separated morphism of algebraic stacks is called **concentrated** if for every base change along a quasi-compact quasi-separated morphism, the derived pushforward has finite cohomological dimension.

2.2.4. *Perfect complexes.* Perfect complexes may be defined on any ringed site [Sta26, Tag 08G4]. This includes the lisse-étale site of $\mathcal{X}$. A complex is **strictly perfect** if it is a bounded complex whose terms are direct summands of finite free modules. A complex is **perfect** if it is locally strictly perfect. By [HR17, Lemmas 4.4 & 4.8], compact objects of $D_{\mathrm{qc}}(\mathcal{X})$ are perfect complexes and the support of a perfect complex has quasi-compact complement.

2.2.5. *Thomason condition.* $\mathcal{X}$ satisfies the β-**Thomason condition**, for some cardinal β, if $D_{\mathrm{qc}}(\mathcal{X})$ is compactly generated by a set of cardinality at most β, and if for every closed subset $Z \subseteq |\mathcal{X}|$ with quasi-compact complement there exists a perfect complex $P \in \mathrm{Perf}(\mathcal{X})$ such that $\mathrm{supp}(P) = Z$.

2.2.6. *Étale neighborhood.* An **étale neighborhood** is a diagram $\mathcal{U} \xrightarrow{j} \mathcal{X} \xleftarrow{f} \mathcal{Y}$ where j is a quasi-compact open immersion, f is concentrated and étale, and its base change along every morphism to $\mathcal{X}$ with image disjoint from $|\mathcal{U}|$ is an isomorphism (cf. [HR23, Definition 1.1]).

3. Auxiliary results

Lemma 3.1. *Let $\mathcal{T}$ and $\mathcal{T}'$ be triangulated categories which admit small coproducts. Choose $H \in \mathcal{T}$ and integers $a \leq b$. Then:*

1. *$\overline{\langle H \rangle}^{[a,b]}$ is closed under small coproducts, extensions, and direct summands; in fact, it is the smallest strictly full subcategory of $\mathcal{T}$ closed under these operations that contains $H[a,b]$.*
2. *$\overline{\langle H \rangle}^{[a,b]}[s] = \overline{\langle H \rangle}^{[a-s,b-s]}$ and $\overline{\langle H \rangle}^{[a,b]} \subseteq \overline{\langle H \rangle}^{[a',b']}$ for all $s \in \mathbb{Z}$, $a' \leq a$, and $b \leq b'$.*
3. *$F(\overline{\langle H \rangle}^{[a,b]}) \subseteq \overline{\langle H' \rangle}^{[a+c,b+d]}$ for any small coproduct preserving exact $F\colon \mathcal{T} \to \mathcal{T}'$ and $H' \in \mathcal{T}'$ with $F(H) \in \overline{\langle H' \rangle}^{[c,d]}$.*
4. *Let $Y \in \mathcal{T}$. If $\mathrm{Hom}(H[-k], Y) = 0$ for all $a \leq k \leq b$, then $\mathrm{Hom}(X, Y) = 0$ for all $X \in \overline{\langle H \rangle}^{[a,b]}$. If H is compact, then $\{Y \in \mathcal{T} : \mathrm{Hom}(H, Y) = 0\}$ is closed under small coproducts, extensions, and direct summands.*
5. *$\mathrm{smd}(\mathrm{Coprod}(H[a,b])) \subseteq \mathrm{Coprod}(H[a,b+1])$.*

Proof. First we prove (1). By definition, $\overline{\langle H \rangle}^{[a,b]} := \mathrm{smd}(\mathrm{Coprod}(H[a,b]))$. Let Λ be a set. Choose $X_\lambda \in \overline{\langle H \rangle}^{[a,b]}$ where $\lambda \in \Lambda$. For each $\lambda \in \Lambda$, there exists $X'_\lambda \in \mathcal{T}$ such that $X_\lambda \oplus X'_\lambda \in \mathrm{Coprod}(H[a,b])$. Since $\mathrm{Coprod}(H[a,b])$ is closed under small coproducts, $\oplus_{\lambda\in\Lambda}(X_\lambda \oplus X'_\lambda) \in \mathrm{Coprod}(H[a,b])$. Note that $\oplus_{\lambda\in\Lambda} X_\lambda$ is a direct summand of $\oplus_{\lambda\in\Lambda}(X_\lambda \oplus X'_\lambda)$. Hence, $\overline{\langle H \rangle}^{[a,b]}$ is closed under small coproducts.

Consider a distinguished triangle

$$A \to B \to C \to A[1]$$

where $A, C \in \overline{\langle H \rangle}^{[a,b]}$. We can find A', C' such that $A \oplus A', C \oplus C' \in \mathrm{Coprod}(H[a,b])$. There exist distinguished triangles

$$A' \xrightarrow{1_{A'}} A' \to 0 \to A'[1]$$

and

$$0 \to C' \xrightarrow{1_{C'}} C' \to 0[1].$$

Taking the direct sum of these distinguished triangles yields a distinguished triangle

$$A \bigoplus A' \bigoplus 0 \to B \bigoplus A' \bigoplus C' \to C \bigoplus 0 \bigoplus C' \to A[1] \bigoplus A'[1] \bigoplus 0.$$

See [Nee01, Proposition 1.2.1]. Since $A \oplus A', C \oplus C' \in \mathrm{Coprod}(H[a,b])$, it follows that $B \oplus A' \oplus C' \in \mathrm{Coprod}(H[a,b])$ because the category is closed under extensions. Consequently, $B \in \overline{\langle H \rangle}^{[a,b]}$.

To see that $\overline{\langle H \rangle}^{[a,b]}$ is closed under direct summands, use the fact that a direct summand of a direct summand remains a direct summand.

Lastly, we show the minimality claim. Let $\mathcal{B}$ be a strictly full subcategory of $\mathcal{T}$ closed under small coproducts, extensions, and direct summands that contains $H[a,b]$. In this case, $\mathrm{Coprod}(H[a,b]) \subseteq \mathcal{B}$. Indeed, $\mathcal{B}$ is closed under small coproducts and extensions, and it contains $H[a,b]$. Moreover, $\mathrm{smd}(\mathrm{Coprod}(H[a,b])) \subseteq \mathcal{B}$ because $\mathcal{B}$ is closed under direct summands.

Second we prove (2). The autoequivalence $[s]\colon \mathcal{T} \to \mathcal{T}$ preserves extensions, small coproducts, and direct summands. Hence, $\overline{\langle H \rangle}^{[a,b]}[s]$ is the smallest strictly full subcategory containing $(H[a,b])[s]$ which is closed under extensions, small coproducts, and direct summands. However, shifting by s yields

$$\begin{aligned}(H[a,b])[s] &= \{H[-k][s] : a \le k \le b\} \\ &= \{H[-k+s] : a \le k \le b\} \\ &= \{H[-l] : a - s \le l \le b - s\} \\ &=: H[a-s, b-s].\end{aligned}$$

It follows that $\overline{\langle H \rangle}^{[a,b]}[s] = \overline{\langle H \rangle}^{[a-s,b-s]}$. Moreover, as $H[a,b] \subseteq H[a',b']$, (1) implies $\overline{\langle H \rangle}^{[a,b]} \subseteq \overline{\langle H \rangle}^{[a',b']}$.

Third we prove (3). Choose $a \le k \le b$. Note that F is exact and $F(H) \in \overline{\langle H' \rangle}^{[c,d]}$. Hence, by (2),

$$F(H[-k]) \cong F(H)[-k] \in \overline{\langle H' \rangle}^{[c,d]}[-k] = \overline{\langle H' \rangle}^{[c+k,d+k]} \subseteq \overline{\langle H' \rangle}^{[a+c,b+d]}.$$

Set $\mathcal{S}$ to be the full subcategory of $E \in \mathcal{T}$ such that $F(E) \in \overline{\langle H' \rangle}^{[a+c,b+d]}$. It follows that $H[a,b] \subseteq \mathcal{S}$. By (1), $\overline{\langle H' \rangle}^{[a+c,b+d]}$ is strictly full and closed under small coproducts, extensions, and direct summands. Since F preserves small coproducts, $\mathcal{S}$ has the same properties. By minimality, (1) implies $\overline{\langle H \rangle}^{[a,b]} \subseteq \mathcal{S}$. Thus, (3) follows.

Next we prove (4). It is easy to see that $\{E : \mathrm{Hom}(E,Y) = 0\}$ is strictly full. Let Λ be a set. Choose $X_\lambda \in \{E : \mathrm{Hom}(E,Y) = 0\}$. By the universal property of coproducts, there

exists an isomorphism

$$\mathrm{Hom}(\bigoplus_{\lambda\in\Lambda} X_\lambda, Y) \cong \prod_{\lambda\in\Lambda} \mathrm{Hom}(X_\lambda, Y) \cong 0.$$

See [Nee01, Example 1.1.13] applied to $\mathcal{T}^{op}$. The last isomorphism follows from the fact that each $X_\lambda \in \{E : \mathrm{Hom}(E,Y) = 0\}$. Hence, $\{E : \mathrm{Hom}(E,Y) = 0\}$ is closed under small coproducts. Let $A, B \in \{E : \mathrm{Hom}(E,Y) = 0\}$. Consider an extension C given by a distinguished triangle

$$A \to C \to B \to A[1].$$

Apply $\mathrm{Hom}(-,Y)$ to obtain a long exact sequence in Hom sets

$$\cdots \to \mathrm{Hom}(B,Y) \to \mathrm{Hom}(C,Y) \to \mathrm{Hom}(A,Y) \to \cdots.$$

Hence, $\mathrm{Hom}(C,Y) \cong 0$ if

$$\mathrm{Hom}(A,Y) \cong \mathrm{Hom}(B,Y) \cong 0.$$

This implies $\{E : \mathrm{Hom}(E,Y) = 0\}$ is closed under extensions. Moreover, if $A \oplus B \in \{E : \mathrm{Hom}(E,Y) = 0\}$, then

$$\mathrm{Hom}(A \oplus B, Y) \cong \mathrm{Hom}(A,Y) \bigoplus \mathrm{Hom}(B,Y) \cong 0.$$

Thus, $\{E : \mathrm{Hom}(E,Y) = 0\}$ is closed under direct summands. The hypothesis asserts that $H[a,b] \subseteq \{E : \mathrm{Hom}(E,Y) = 0\}$. By (1), $\overline{\langle H\rangle}^{[a,b]} \subseteq \{E : \mathrm{Hom}(E,Y) = 0\}$. A similar argument with H compact proves the second claim for (4). Indeed, the compactness of H gives the closure under coproducts, whereas the other conditions are analogous.

Lastly we prove (5). Let $X \in \mathrm{smd}(\mathrm{Coprod}(H[a,b]))$. Choose X' such that $X \oplus X' \in \mathrm{Coprod}(H[a,b])$. The so-called Eilenberg swindle yields an isomorphism

$$\begin{aligned} X \oplus (\bigoplus_{n\in\mathbb{N}} (X \oplus X')) &\cong X \oplus (\bigoplus_{n\in\mathbb{N}} (X' \oplus X)) \\ &\cong X \oplus (X' \oplus X) \oplus (X' \oplus X) \oplus \cdots \\ &\cong (X \oplus X') \oplus (X \oplus X') \oplus \cdots \\ &\cong \bigoplus_{n\in\mathbb{N}} (X \oplus X'). \end{aligned}$$

Consequently, there exists a distinguished triangle

$$X \to X \oplus (\bigoplus_{n\in\mathbb{N}} (X \oplus X')) \to \bigoplus_{n\in\mathbb{N}} (X \oplus X') \to X[1].$$

Rotating backwards yields

$$(\bigoplus_{n\in\mathbb{N}} (X \oplus X'))[-1] \to X \to X \oplus (\bigoplus_{n\in\mathbb{N}} (X \oplus X')) \to \bigoplus_{n\in\mathbb{N}} (X \oplus X').$$

Since $[a,b] \subseteq [a,b+1]$, $\mathrm{Coprod}(H[a,b]) \subseteq \mathrm{Coprod}(H[a,b+1])$. Observe that

$$\begin{aligned} (\bigoplus_{n\in\mathbb{N}} (X \oplus X'))[-1] &\in \mathrm{Coprod}(H[a,b])[-1] \\ &= \mathrm{Coprod}(H[a+1,b+1]) \\ &\subseteq \mathrm{Coprod}(H[a,b+1]). \end{aligned}$$

Therefore,

$$\begin{aligned} X &\in \mathrm{Coprod}(H[a,b+1]) \star \mathrm{Coprod}(H[a,b]) \\ &\subseteq \mathrm{Coprod}(H[a,b+1]) \star \mathrm{Coprod}(H[a,b+1]) \\ &\subseteq \mathrm{Coprod}(H[a,b+1]). \end{aligned}$$

This completes the proof. □

Remark 3.2. We use the octahedral axiom in the following ways. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be composable.

(1) Choose distinguished triangles $X \xrightarrow{f} Y \to C_f \to X[1]$, $Y \xrightarrow{g} Z \xrightarrow{p_g} C_g \to Y[1]$ and $X \xrightarrow{gf} Z \xrightarrow{p_{gf}} C_{gf} \to X[1]$. There is a distinguished triangle $C_f \to C_{gf} \xrightarrow{h} C_g \to C_f[1]$ with $h \circ p_{gf} = p_g$. See e.g. [Nee01, Proposition 1.4.6].

(2) Choose distinguished triangles $F_f \to X \xrightarrow{f} Y \to F_f[1]$, $F_g \to Y \xrightarrow{g} Z \to F_g[1]$ and $F_{gf} \to X \xrightarrow{gf} Z \to F_{gf}[1]$. There is a distinguished triangle $F_f \to F_{gf} \to F_g \to F_f[1]$. Indeed, take the distinguished triangle $C_f \to C_{gf} \xrightarrow{h} C_g \to C_f[1]$ and shift by $[-1]$. Here we use that $F_? \cong C_?[-1]$.

Lemma 3.3. *Let $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0})$ be a t-structure on a triangulated category $\mathcal{T}$. Consider a distinguished triangle*

$$E \to F \to D \to E[1]$$

where $F \in \mathcal{T}^{\leq 0}$ and $D \in \mathcal{T}^{\leq -m}$ for some $m \geq 1$. Then $E \in \mathcal{T}^{\leq 0}$.

Proof. By rotating backwards, there exists a distinguished triangle

$$D[-1] \to E \to F \to D.$$

Note that $D[-1] \in \mathcal{T}^{\leq 1-m} \subseteq \mathcal{T}^{\leq 0}$. Since aisles are closed under extensions, $E \in \mathcal{T}^{\leq 0}$. □

Lemma 3.4. *Let $(H, \mathcal{T}^{\leq 0}, A)$ be weak approximation data for $\mathcal{T}$. For every $F \in \mathcal{T}^{\leq 0}$ and every $m \geq 1$, there exists a distinguished triangle*

$$E_m \to F \to D_m \to E_m[1]$$

with $E_m \in \overline{\langle H \rangle}^{[1-m-A,A]} \cap \mathcal{T}^{\leq 0}$ and $D_m \in \mathcal{T}^{\leq -m}$.

Proof. This follows from [Nee26, Corollary 3.2(1)] with weak approximation data for $\mathcal{T}$. To be self-contained, we give a proof. By Lemma 3.3, for any desired distinguished triangle

$$E_m \to F \to D_m \to E_m[1],$$

it follows that $E_m \in \mathcal{T}^{\leq 0}$. Hence, it suffices to produce the desired distinguished triangle. We argue by induction on m. The base case $m = 1$ is Definition 2.1(2).

Assume that the desired distinguished triangle exists for m, i.e. a distinguished triangle

$$E_m \to F \xrightarrow{f} D_m \to E_m[1]$$

with $E_m \in \overline{\langle H \rangle}^{[1-m-A,A]} \cap \mathcal{T}^{\leq 0}$ and $D_m \in \mathcal{T}^{\leq -m}$. Since $D_m[-m] \in \mathcal{T}^{\leq 0}$, Definition 2.1(2) yields a distinguished triangle

$$E' \to D_m[-m] \to D' \to E'[1]$$

where $E' \in \overline{\langle H \rangle}^{[-A,A]}$ and $D' \in \mathcal{T}^{\leq -1}$. After shifting by $[m]$, we have a distinguished triangle

$$E'[m] \to D_m \xrightarrow{g} D'[m] \to E'[m+1]$$

where $E'[m] \in \overline{\langle H \rangle}^{[-A-m,A-m]}$ and $D'[m] \in \mathcal{T}^{\leq -1-m}$. Set $D_{m+1} := D'[m]$. Complete the composite $g \circ f : F \to D'[m]$ to a distinguished triangle,

$$F \to D'[m] \to R \to F[1].$$

Set $E_{m+1} := R[-1]$. The cocones of f, g, and $g \circ f$ are respectively E_m, $E'[m]$, and E_{m+1}. By Remark 3.2(2), this yields a distinguished triangle

$$E_m \to E_{m+1} \to E'[m] \to E_m[1].$$

By Lemma 3.1(1)(2),

$$\begin{aligned} E_{m+1} &\in \overline{\langle H \rangle}^{[1-m-A,A]} \star \overline{\langle H \rangle}^{[-m-A,A-m]} \\ &\subseteq \overline{\langle H \rangle}^{[-m-A,A]} \\ &= \overline{\langle H \rangle}^{[1-(m+1)-A,A]}. \end{aligned}$$

This completes the proof. □

Lemma 3.5. *Let $\mathcal{T}$ be a triangulated category admitting small coproducts. Choose $H \in \mathcal{T}^c$. Denote by $\mathcal{T}^{\leq 0}$ the aisle compactly generated by H. Assume that* $\mathrm{Hom}(H, H[n]) = 0$ *for all $n \geq n_0$. Then* $\mathrm{Hom}(H, \mathcal{T}^{\leq -n_0}) = 0$.

Proof. This is [Nee26, Remark 1.24]; cf. [BNP23, Lemma 3.5(1)]. On one hand,

$$\begin{aligned} \mathcal{T}^{\leq -n_0} &= \mathcal{T}^{\leq 0}[n_0] \\ &= \mathrm{Coprod}(\{H[n] : n \geq n_0\}). \end{aligned}$$

On the other hand, the strictly full subcategory

$$\{Y \in \mathcal{T} : \mathrm{Hom}(H, Y) = 0\}$$

contains $H[n]$ for $n \geq n_0$. Moreover, by Lemma 3.1(4), this subcategory above is closed under small coproducts and extensions. Consequently, the subcategory above contains $\mathcal{T}^{\leq -n_0}$. Thus, the desired claim follows. □

4. Lifts to aisles

Setup 4.1. Consider a recollement of singly compactly generated triangulated categories:

$$\mathcal{D}_Z \underset{i^!}{\overset{i^*}{\rightleftarrows}} \underset{i_*}{\mathcal{D}} \underset{j_*}{\overset{j_!}{\rightleftarrows}} \mathcal{D}_U.$$

For every $E \in \mathcal{D}$, there are distinguished triangles

$$\begin{aligned} &j_! j^* E \xrightarrow{\epsilon_E} E \xrightarrow{\eta_E} i_* i^* E \xrightarrow{\varphi_E} j_! j^* E[1], \\ &i_* i^! E \to E \to j_* j^* E \to i_* i^! E[1]. \end{aligned} \tag{4.1}$$

Here ϵ is the counit of $j_! \dashv j^*$ and η is the unit of $i^* \dashv i_*$.

Lemma 4.2. *Assume Setup 4.1. Consider t-structures* $(\mathcal{D}_Z^{\leq 0}, \mathcal{D}_Z^{\geq 0})$ *and* $(\mathcal{D}_U^{\leq 0}, \mathcal{D}_U^{\geq 0})$ *respectively on* $\mathcal{D}_Z$ *and* $\mathcal{D}_U$*. Denote by* $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ *the glued t-structure. For every* $n \in \mathbb{Z}$*,*

$$\mathcal{D}^{\leq n} = \{E \in \mathcal{D} : \ i^*E \in \mathcal{D}_Z^{\leq n} \text{ and } j^*E \in \mathcal{D}_U^{\leq n}\}.$$

In particular,

$$i^*\mathcal{D}^{\leq n} \subseteq \mathcal{D}_Z^{\leq n}, \qquad j^*\mathcal{D}^{\leq n} \subseteq \mathcal{D}_U^{\leq n}, \qquad i_*\mathcal{D}_Z^{\leq n} \subseteq \mathcal{D}^{\leq n}, \qquad j_!\mathcal{D}_U^{\leq n} \subseteq \mathcal{D}^{\leq n}.$$

Proof. If $n = 0$, then this is the definition of the glued aisle. Since i^* and j^* commute with shifts,

$$\mathcal{D}^{\leq n} = \{E \in \mathcal{D} : \ i^*E \in \mathcal{D}_Z^{\leq n} \text{ and } j^*E \in \mathcal{D}_U^{\leq n}\}.$$

Moreover, the functors $i^*, j^*, i_*, j_!$ are right t-exact for the $n = 0$ case. After shifting, the desired containments follow. □

Lemma 4.3. *Assume Setup 4.1 where* $\mathcal{D}_U$ *is weakly approximable with compact generator* G_U*. Suppose* G' *is a compact generator for* $\mathcal{D}$ *which satisfies* $\mathrm{Hom}(G', G'[n]) = 0$ *for* $n \gg 0$*. Denote by* $\mathcal{D}_U^{\leq 0}$ *the aisle compactly generated by* G_U*. Then the following hold:*

(1) there exists an $m \geq 0$ *such that* $j^*G' \in \mathcal{D}_U^{\leq m}$
(2) $G := G'[m] \oplus j_!G_U$ *is a compact generator of* $\mathcal{D}$*, and* i^*G *is a compact generator of* $\mathcal{D}_Z$
(3) the t-structure on $\mathcal{D}$ *compactly generated by* G *is the gluing of the t-structure on* $\mathcal{D}_Z$ *compactly generated by* i^*G *with* $(\mathcal{D}_U^{\leq 0}, \mathcal{D}_U^{\geq 0})$
(4) $\mathrm{Hom}(G, G[n]) = 0$ *for* $n \gg 0$*.*

Proof. (1) and (3) are essentially from [BNP23, Lemma 3.11]. The observation below is [BNP23, Lemma 3.5(2)] (see also [GL26, Lemma A.1]). We recall the argument for convenience.

We prove (1). Recall that $j_!$ preserves compact objects. Hence, $j_!G_U \in \mathcal{D}^c$. We observe that for all $F, F' \in \mathcal{D}^c$, there exists an integer N such that $\mathrm{Hom}(F, F'[n]) = 0$ for all $n \geq N$. First, consider the full subcategory $\mathcal{H}$ of $F \in \mathcal{D}^c$ where such an N exists with $F' = G'$. This subcategory is thick (e.g. use the long exact sequences of $\mathrm{Hom}(-, G'[n])$). By hypothesis, $G' \in \mathcal{H}$. As G' is a classical generator of $\mathcal{D}^c$ (see §2.1.2), $\mathcal{H} = \mathcal{D}^c$. Second, fix $F \in \mathcal{D}^c$. Consider the full subcategory $\mathcal{H}'$ of $F' \in \mathcal{D}^c$ for which such an N exists. Again, $\mathcal{H}'$ is thick, which can be shown using the long exact sequences of $\mathrm{Hom}(F, -)$. Also, by the first step above, $G' \in \mathcal{H}'$. Hence, $\mathcal{H}' = \mathcal{D}^c$, which proves the observation. Applying the observation to $F = j_!G_U$ and $F' = G'$, and using adjunction, we have for $n \gg 0$,

$$\mathrm{Hom}(G_U[-n], j^*G') \cong \mathrm{Hom}(j_!G_U[-n], G') = 0.$$

Since $\mathcal{D}_U$ is weakly approximable and $\mathcal{D}_U^{\leq 0}$ is in the preferred equivalence class, [BNP23, Corollary 3.10(1)] implies that $j^*G' \in \mathcal{D}_U^{\leq m}$ for some $m \geq 0$.

Note that $j_!G_U$ is compact. Hence, the hypotheses of [GL26, Lemma A.1] hold: G' and G_U are respectively compact generators of $\mathcal{D}$ and $\mathcal{D}_U$, $\mathcal{D}_U^{\leq 0}$ is the aisle compactly generated by G_U, and $j^*G' \in \mathcal{D}_U^{\leq m}$ by (1). Consequently, (2) and (3) follow from [GL26, Lemma A.1].

To prove (4), apply the observation with $F = F' = G$. □

Setup 4.4. Consider Setup 4.1. Assume that $\mathcal{D}_Z$ and $\mathcal{D}_U$ are weakly approximable. Suppose $\mathcal{D}$ has a compact generator G' satisfying $\mathrm{Hom}(G', G'[n]) = 0$ for $n \gg 0$. Fix a compact generator G_U of $\mathcal{D}_U$. Choose m and $G = G'[m] \oplus j_!G_U$ from Lemma 4.3. Consider the following t-structures:

- $\mathcal{D}_U^{\leq 0}$ is compactly generated by G_U;
- $\mathcal{D}_Z^{\leq 0}$ is compactly generated by i^*G;
- $\mathcal{D}^{\leq 0}$ is the glued t-structure (by Lemma 4.3(3), it is the t-structure compactly generated by G; hence, $G \in \mathcal{D}^{\leq 0}$).

Note that $\mathcal{D}_U^{\leq 0}$ and $\mathcal{D}_Z^{\leq 0}$ are respectively in the preferred equivalence classes of $\mathcal{D}_U$ and $\mathcal{D}_Z$. By [BNP23, Lemma 3.7] (equivalently, [Nee26, Proposition 3.6]), Lemma 3.5, and Lemma 4.3(4), there exist $A_1, A_2, A_3 \gg 0$ such that the following hold:

(1) $(i^*G, \mathcal{D}_Z^{\leq 0}, A_1)$ is weak approximation data for $\mathcal{D}_Z$
(2) $(G_U, \mathcal{D}_U^{\leq 0}, A_2)$ is weak approximation data for $\mathcal{D}_U$
(3) $\mathrm{Hom}(G, \mathcal{D}^{\leq -A_3}) = 0$.

Set $A = \max\{A_1, A_2, A_3\}$. Define $K := A + 1$, $B := 3A$, and $\mathcal{P} := \overline{\langle G \rangle}^{[-B,B]} \cap \mathcal{D}^{\leq K}$. By Lemma 3.1(1), and the fact that aisles are closed under extensions and small coproducts, $\mathcal{P}$ is closed under small coproducts and extensions. Since $j_!G_U$ is a direct summand of G, it follows that $j_!G_U \in \overline{\langle G \rangle}^{[0,0]}$.

Definition 4.5. Assume Setup 4.4. Let $E' \in \mathcal{D}_Z$, $Y \in \mathcal{D}$, and $f' \colon E' \to i^*Y$. A $\mathcal{P}$-**lift** of f' is a pair (q, θ) consisting of a morphism $q \colon E \to Y$ with $E \in \mathcal{P}$ and an isomorphism $\theta \colon i^*E \to E'$ such that $i^*q = f' \circ \theta$.

Lemma 4.6. *Assume Setup 4.4. Let $Y \in \mathcal{D}^{\leq K}$, $-A - 1 \leq n \leq A$, and $f' \colon (i^*G)[n] \to i^*Y$. Then f' admits a $\mathcal{P}$-lift.*

Proof. The following proof is a variant of [BNP23, Lemma 5.3]. Specifically, the lift is constructed directly from the adjunct g rather than from a roof [BNP23, Lemma 5.2], and it lies in $\mathcal{D}^{\leq K}$ for the same K that bounds Y.

Since i^* is exact, $i^*(G[n])$ can be identified with $(i^*G)[n]$. Hence, f' yields a morphism $i^*(G[n]) \to i^*Y$. Denote by η the unit and $\overline{\varepsilon}$ the counit of $i^* \dashv i_*$.

Set $M := j^*Y$. By Lemma 4.2, $M \in \mathcal{D}_U^{\leq K}$, and so $M[K] \in \mathcal{D}_U^{\leq 0}$. Apply Lemma 3.4 to the weak approximation data $(G_U, \mathcal{D}_U^{\leq 0}, A)$ of Setup 4.4(2), $M[K]$, and $3A + 1 = K + 2A$ in place of m. This gives a distinguished triangle

$$W' \to M[K] \to V' \to W'[1].$$

Here $W' \in \overline{\langle G_U \rangle}^{[-4A,A]} \cap \mathcal{D}_U^{\leq 0}$ and $V' \in \mathcal{D}_U^{\leq -3A-1}$. Define $W := W'[-K]$ and $V := V'[-K]$. Shifting by $[-K]$ yields a distinguished triangle

$$W \xrightarrow{w} M \xrightarrow{c} V \to W[1]. \tag{4.2}$$

By Lemma 3.1(2),

$$W \in \overline{\langle G_U \rangle}^{[-4A+K, A+K]} = \overline{\langle G_U \rangle}^{[1-3A, 2A+1]}.$$

Moreover, $W \in \mathcal{D}_U^{\leq 0}[-K] = \mathcal{D}_U^{\leq K}$ and $V \in \mathcal{D}_U^{\leq -3A-1}[-K] = \mathcal{D}_U^{\leq -2A}$.

Now let $g \colon G[n] \to i_*i^*Y$ be the adjunct of f'. That is, $g = i_*(f') \circ \eta_{G[n]}$, and so $f' = \overline{\varepsilon}_{i^*Y} \circ i^*(g)$. Set

$$\varphi := \varphi_Y \colon i_*i^*Y \to j_!M[1]$$

to be the morphism in (4.1), and $\xi := \varphi \circ g$. By Lemma 4.2, $j_!V \in \mathcal{D}^{\leq -2A}$. As $n \leq A$,

$$j_!V[1-n] \in \mathcal{D}^{\leq n-1-2A} \subseteq \mathcal{D}^{\leq -A}.$$

Hence, Setup 4.4(3) gives

$$\operatorname{Hom}(G[n], j_!V[1]) \cong \operatorname{Hom}(G, j_!V[1-n]) = 0. \tag{4.3}$$

Applying $j_!$ to (4.2) gives a distinguished triangle

$$j_!W \xrightarrow{j_!(w)} j_!M \xrightarrow{j_!(c)} j_!V \to j_!W[1].$$

By [Nee01, Lemma 1.1.10], $\operatorname{Hom}(G[n-1], -)$ is a homological functor on $\mathcal{D}$. This yields an exact sequence

$$\operatorname{Hom}(G[n-1], j_!W) \xrightarrow{j_!(w)\circ -} \operatorname{Hom}(G[n-1], j_!M) \xrightarrow{j_!(c)\circ -} \operatorname{Hom}(G[n-1], j_!V).$$

By (4.3),

$$\operatorname{Hom}(G[n-1], j_!V) \cong \operatorname{Hom}(G[n], j_!V[1]) = 0.$$

Hence, the left-hand map is surjective. Thus, there exists $\xi'' \colon G[n-1] \to j_!W$ such that $\xi[-1] = j_!(w) \circ \xi''$. Setting $\xi' := \xi''[1] \colon G[n] \to j_!W[1]$, we get $\xi = j_!(w)[1] \circ \xi'$.

Next we construct the lift. Completing ξ' and rotating backwards twice yields a distinguished triangle

$$j_!W \xrightarrow{\alpha} Q \xrightarrow{\beta} G[n] \xrightarrow{\xi'} j_!W[1]. \tag{4.4}$$

Consider the diagram

$$\begin{array}{ccccccc} j_!W & \xrightarrow{\alpha} & Q & \xrightarrow{\beta} & G[n] & \xrightarrow{\xi'} & j_!W[1] \\ {\scriptstyle j_!(w)}\downarrow & & \vdots{\scriptstyle q} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle j_!(w)[1]} \\ j_!M & \xrightarrow{\epsilon_Y} & Y & \xrightarrow{\eta_Y} & i_*i^*Y & \xrightarrow{\varphi} & j_!M[1] \end{array}$$

with distinguished rows. Indeed, the top row is (4.4), whereas the bottom row is the first triangle of (4.1) for $E = Y$ (i.e. use that $M = j^*Y$). The right-hand square commutes because

$$\varphi \circ g = \xi = j_!(w)[1] \circ \xi'.$$

By TR3, applied after rotating both rows backwards, there exists $q \colon Q \to Y$ making the diagram commute. In particular, $\eta_Y \circ q = g \circ \beta$.

Set $\theta := i^*\beta \colon i^*Q \to i^*(G[n]) = (i^*G)[n]$. Recall that i^* is exact and $i^*j_! = 0$. Applying i^* to (4.4) gives a distinguished triangle

$$0 \to i^*Q \xrightarrow{\theta} i^*(G[n]) \to 0[1],$$

and so θ is an isomorphism. Moreover, applying i^* to $\eta_Y \circ q = g \circ \beta$ and using the triangle identity $\overline{\varepsilon}_{i^*Y} \circ i^*(\eta_Y) = 1_{i^*Y}$, it follows that

$$i^*q = \overline{\varepsilon}_{i^*Y} \circ i^*(\eta_Y) \circ i^*q = \overline{\varepsilon}_{i^*Y} \circ i^*(g) \circ i^*\beta = f' \circ \theta.$$

Lastly, we check that Q lies in $\mathcal{P}$. By (4.4), $Q \in j_!W \star G[n]$. Since $\mathcal{P}$ is closed under extensions, it suffices to show that $j_!W, G[n] \in \mathcal{P}$. The functor $j_!$ is exact, preserves small coproducts, and satisfies $j_!G_U \in \overline{\langle G \rangle}^{[0,0]}$ by Setup 4.4. By Lemma 3.1(3), $j_!W \in \overline{\langle G \rangle}^{[1-3A,2A+1]}$. As $A \geq 1$, we have $2A+1 \leq 3A = B$, so Lemma 3.1(2) gives $j_!W \in \overline{\langle G \rangle}^{[-B,B]}$. Moreover, by Lemma 4.2, $W \in \mathcal{D}_U^{\leq K}$ implies $j_!W \in \mathcal{D}^{\leq K}$. Since $-A-1 \leq n \leq A$, one has $-A \leq -n \leq A+1$, and so $G[n] = G[-(-n)] \in G[-A, A+1]$. Applying Lemma 3.1(2) with

$A+1 \leq 3A$, $G[n] \in \overline{\langle G \rangle}^{[-A,A+1]} \subseteq \overline{\langle G \rangle}^{[-B,B]}$. As $G \in \mathcal{D}^{\leq 0}$ by Setup 4.4, $G[n] \in \mathcal{D}^{\leq -n} \subseteq \mathcal{D}^{\leq A+1} = \mathcal{D}^{\leq K}$. Thus, $Q \in \mathcal{P}$, and so (q,θ) is a $\mathcal{P}$-lift of f'. □

Notation 4.7. Assume Setup 4.4. Denote by $\mathcal{C} \subseteq \mathcal{D}_Z$ the full subcategory of objects E' such that every morphism $f' \colon E' \to i^*Y$ with $Y \in \mathcal{D}^{\leq K}$ admits a $\mathcal{P}$-lift.

Proposition 4.8. *Assume Setup 4.4. The subcategory $\mathcal{C}$ is strictly full, closed under small coproducts and extensions, and contains $(i^*G)[n]$ for $-A-1 \leq n \leq A$. Consequently,*

$$\overline{\langle i^*G \rangle}^{[-A,A]} \subseteq \operatorname{Coprod}((i^*G)[-A,A+1]) \subseteq \mathcal{C}.$$

Proof. First we show that $\mathcal{C}$ is strictly full. Let $\sigma \colon E' \to E''$ be an isomorphism with $E'' \in \mathcal{C}$. Fix $f' \colon E' \to i^*Y$ with $Y \in \mathcal{D}^{\leq K}$. Since $E'' \in \mathcal{C}$, $f' \circ \sigma^{-1}$ has a $\mathcal{P}$-lift (q,θ). Then $\sigma^{-1} \circ \theta$ is an isomorphism, and $i^*q = f' \circ (\sigma^{-1} \circ \theta)$. Thus, $(q, \sigma^{-1} \circ \theta)$ is a $\mathcal{P}$-lift of f'. Hence, $E' \in \mathcal{C}$.

By Lemma 4.6, $(i^*G)[n] \in \mathcal{C}$ for $-A-1 \leq n \leq A$.

Next we show that $\mathcal{C}$ is closed under small coproducts. Choose a set Λ. Pick some $E'_\lambda \in \mathcal{C}$ for each $\lambda \in \Lambda$. Put $E' := \bigoplus_{\lambda \in \Lambda} E'_\lambda$. Denote by $\iota_\lambda \colon E'_\lambda \to E'$ the inclusions. Fix $f' \colon E' \to i^*Y$ such that $Y \in \mathcal{D}^{\leq K}$. For each $\lambda \in \Lambda$, there exists a $\mathcal{P}$-lift $(q_\lambda \colon E_\lambda \to Y, \theta_\lambda)$ of $f' \circ \iota_\lambda$. Define $E := \bigoplus_{\lambda \in \Lambda} E_\lambda$. Set $\kappa_\lambda \colon E_\lambda \to E$ to be the inclusions for each $\lambda \in \Lambda$. Since $\mathcal{P}$ is closed under small coproducts, $E \in \mathcal{P}$. Denote by $q \colon E \to Y$ the morphism with $q \circ \kappa_\lambda = q_\lambda$ for each $\lambda \in \Lambda$. As i^* preserves small coproducts, i^*E is the coproduct of the i^*E_λ with inclusions $i^*(\kappa_\lambda)$ for $\lambda \in \Lambda$. Hence, there exists $\theta \colon i^*E \to E'$ with $\theta \circ i^*(\kappa_\lambda) = \iota_\lambda \circ \theta_\lambda$ for each $\lambda \in \Lambda$. Note that θ is an isomorphism because it is a coproduct of the isomorphisms θ_λ. For each $\lambda \in \Lambda$,

$$i^*q \circ i^*(\kappa_\lambda) = i^*(q_\lambda) = f' \circ \iota_\lambda \circ \theta_\lambda = f' \circ \theta \circ i^*(\kappa_\lambda).$$

Hence, by the universal property of coproducts, $i^*q = f' \circ \theta$. Thus, (q,θ) is a $\mathcal{P}$-lift of f'.

Now we prove that $\mathcal{C}$ is closed under extensions. Consider a distinguished triangle

$$E'_1 \xrightarrow{u'} E' \to E'_2 \to E'_1[1]$$

where $E'_1, E'_2 \in \mathcal{C}$. Fix $f' \colon E' \to i^*Y$ with $Y \in \mathcal{D}^{\leq K}$. We will lift f' in two stages: first on E'_1, and then on E'_2, into the cone of the first lift.

We start with E'_1. As $E'_1 \in \mathcal{C}$, $f'_1 := f' \circ u'$ has a $\mathcal{P}$-lift $(q_1 \colon E_1 \to Y, \theta_1)$. Complete to a distinguished triangle

$$E_1 \xrightarrow{q_1} Y \xrightarrow{r} Y' \xrightarrow{s} E_1[1].$$

Since $E_1 \in \mathcal{P} \subseteq \mathcal{D}^{\leq K}$, $E_1[1] \in \mathcal{D}^{\leq K-1} \subseteq \mathcal{D}^{\leq K}$. As aisles are closed under extensions, $Y' \in Y \star E_1[1] \subseteq \mathcal{D}^{\leq K}$. Applying i^* to this distinguished triangle gives the top row of the diagram

$$\begin{array}{ccccccc}
i^*E_1 & \xrightarrow{i^*q_1} & i^*Y & \xrightarrow{i^*r} & i^*Y' & \xrightarrow{i^*s} & (i^*E_1)[1] \\
{\scriptstyle\theta_1}\downarrow & & \| & & \| & & \downarrow{\scriptstyle\theta_1[1]} \\
E'_1 & \xrightarrow{f'_1} & i^*Y & \xrightarrow{i^*r} & i^*Y' & \xrightarrow{\theta_1[1]\circ i^*s} & E'_1[1].
\end{array}$$

Since $i^*q_1 = f'_1 \circ \theta_1$, the left square commutes. By construction, the other two squares commute. As all vertical maps are isomorphisms, the bottom row is isomorphic to a distinguished triangle, and hence is distinguished.

Next we find the morphism that f' induces on E_2'. Again, we complete to a distinguished triangle

$$E' \xrightarrow{f'} i^*Y \xrightarrow{p'} R' \to E'[1].$$

We apply Remark 3.2(1) to $E_1' \xrightarrow{u'} E' \xrightarrow{f'} i^*Y$. For this, we use the given distinguished triangle on u', the chosen distinguished triangle on f', and the bottom row of the diagram above as the distinguished triangle on $f' \circ u' = f_1'$. This gives a distinguished triangle

$$E_2' \xrightarrow{f_2'} i^*Y' \xrightarrow{h} R' \to E_2'[1]$$

such that the following diagram commutes:

$$\begin{array}{ccc} i^*Y & \xrightarrow{i^*r} & i^*Y' \\ & \searrow^{p'} & \downarrow{\scriptstyle h} \\ & & R'. \end{array}$$

Now we lift f_2'. Since $E_2' \in \mathcal{C}$ and $Y' \in \mathcal{D}^{\leq K}$, f_2' has a $\mathcal{P}$-lift $(q_2 \colon E_2 \to Y', \theta_2)$. Choose a distinguished triangle

$$E_2 \xrightarrow{q_2} Y' \xrightarrow{s_2} Y'' \to E_2[1].$$

Next we combine the two lifts. Complete to a distinguished triangle

$$E \xrightarrow{q} Y \xrightarrow{s_2 \circ r} Y'' \to E[1].$$

The cocones of r, s_2, and $s_2 \circ r$ are respectively E_1, E_2, and E. Hence, Remark 3.2(2) gives a distinguished triangle $E_1 \to E \to E_2 \to E_1[1]$. Since $\mathcal{P}$ is closed under extensions, $E \in E_1 \star E_2 \subseteq \mathcal{P}$.

It remains to compare i^*q with f'. We do this in two steps. First, applying i^* to the distinguished triangle on q_2 gives the top row of the diagram

$$\begin{array}{ccccccc} i^*E_2 & \xrightarrow{i^*q_2} & i^*Y' & \xrightarrow{i^*s_2} & i^*Y'' & \longrightarrow & (i^*E_2)[1] \\ {\scriptstyle\theta_2}\downarrow & & \| & & \downarrow{\scriptstyle\chi} & & \downarrow{\scriptstyle\theta_2[1]} \\ E_2' & \xrightarrow{f_2'} & i^*Y' & \xrightarrow{h} & R' & \longrightarrow & E_2'[1]. \end{array}$$

Note that the bottom row is the distinguished triangle on f_2' found above. As $i^*q_2 = f_2' \circ \theta_2$, the left square commutes. By TR3, there exists a morphism $\chi \colon i^*Y'' \to R'$ making the diagram commute. Hence, $\chi \circ i^*s_2 = h$. Since θ_2 and the identity are isomorphisms, so is χ. Combine with $h \circ i^*r = p'$ to obtain

$$\chi \circ i^*(s_2 \circ r) = h \circ i^*r = p'.$$

Applying i^* to the distinguished triangle on q gives

$$\begin{array}{ccccccc} i^*E & \xrightarrow{i^*q} & i^*Y & \xrightarrow{i^*(s_2\circ r)} & i^*Y'' & \longrightarrow & (i^*E)[1] \\ {\scriptstyle\theta}\downarrow & & \| & & \downarrow{\scriptstyle\chi} & & \downarrow{\scriptstyle\theta[1]} \\ E' & \xrightarrow{f'} & i^*Y & \xrightarrow{p'} & R' & \longrightarrow & E'[1]. \end{array}$$

The bottom row is the distinguished triangle on f', whereas the middle square commutes by the diagram above. By TR3, applied after rotating both rows, there exists a morphism

$\theta\colon i^*E \to E'$ making the diagram commute. Hence, $f' \circ \theta = i^*q$. As before, θ is an isomorphism, because 1 and χ are. Since $E \in \mathcal{P}$, (q,θ) is a $\mathcal{P}$-lift of f'. Hence, $E' \in \mathcal{C}$.

Finally, we prove the two inclusions. The objects of $(i^*G)[-A, A+1]$ are $(i^*G)[-k]$ with $-A \le k \le A+1$. For $n = -k$, these are the objects $(i^*G)[n]$ with $-A-1 \le n \le A$. Hence, by Lemma 4.6, these lie in $\mathcal{C}$. Moreover, by definition, $\operatorname{Coprod}((i^*G)[-A, A+1])$ is the smallest strictly full subcategory containing these objects that is closed under small coproducts and extensions. Since $\mathcal{C}$ has all these properties, $\operatorname{Coprod}((i^*G)[-A, A+1]) \subseteq \mathcal{C}$. The first inclusion is Lemma 3.1(5). □

Remark 4.9. Proposition 4.8 replaces the induction on the number of cones in [BNP23, Lemmas 5.4 and 5.5], which requires approximability.

5. Proofs of results

Proof of Theorem 1.1. First, let $\mathcal{D}$ be weakly approximable. Denote by $\mathcal{D}^{\le 0}$ the aisle compactly generated by G'. Note that $G' \in \mathcal{D}^{\le 0}$. By [BNP23, Lemma 3.7], there exists an integer $A' > 0$ such that $(G', \mathcal{D}^{\le 0}, A')$ satisfies Definition 2.1(1). It follows that $\operatorname{Hom}(G', \mathcal{D}^{\le -A'}) = 0$. For $n \ge A'$, $G'[n] \in \mathcal{D}^{\le -n} \subseteq \mathcal{D}^{\le -A'}$. Hence, for all $n \ge A'$, $\operatorname{Hom}(G', G'[n]) = 0$. This argument applies to every compact generator of $\mathcal{D}$.

Conversely, let $\operatorname{Hom}(G', G'[n]) = 0$ for $n \gg 0$. Hence, we are in the setting of Setup 4.4, and we use its notation. By Lemma 4.3(2), G is a compact generator of $\mathcal{D}$. We show that $(G, \mathcal{D}^{\le 0}, 3A)$ is weak approximation data for $\mathcal{D}$.

We start with Definition 2.1(1). Since $G \in \mathcal{D}^{\le 0}$, $G \in \mathcal{D}^{\le 3A}$. As $\mathcal{D}^{\le -3A} \subseteq \mathcal{D}^{\le -A}$, Setup 4.4(3) implies $\operatorname{Hom}(G, \mathcal{D}^{\le -3A}) = 0$.

Now we verify Definition 2.1(2). Fix $X \in \mathcal{D}^{\le 0}$. We will approximate X in two stages: first its image i^*X in $\mathcal{D}_Z$, and then what remains in $\mathcal{D}_U$.

We start on $\mathcal{D}_Z$. By Lemma 4.2, $i^*X \in \mathcal{D}_Z^{\le 0}$. By Setup 4.4(1), there exists a distinguished triangle

$$E' \xrightarrow{f'} i^*X \to D' \to E'[1]$$

where $E' \in \overline{\langle i^*G \rangle}^{[-A,A]}$ and $D' \in \mathcal{D}_Z^{\le -1}$. By Proposition 4.8, $E' \in \mathcal{C}$. Since $X \in \mathcal{D}^{\le 0} \subseteq \mathcal{D}^{\le K}$, f' has a $\mathcal{P}$-lift $(q_1\colon E_1 \to X, \theta)$. Complete to a distinguished triangle

$$E_1 \xrightarrow{q_1} X \xrightarrow{r} N \to E_1[1].$$

Since $E_1 \in \mathcal{P} \subseteq \mathcal{D}^{\le K}$, $E_1[1] \in \mathcal{D}^{\le K-1} \subseteq \mathcal{D}^{\le K}$. As aisles are closed under extensions, $N \in X \star E_1[1] \subseteq \mathcal{D}^{\le K}$. Applying i^* to the distinguished triangle on q_1 gives the top row of the diagram

$$\begin{array}{ccccccc}
i^*E_1 & \xrightarrow{i^*q_1} & i^*X & \xrightarrow{i^*r} & i^*N & \longrightarrow & (i^*E_1)[1] \\
\downarrow{\scriptstyle\theta} & & \| & & \downarrow{\scriptstyle\mu} & & \downarrow{\scriptstyle\theta[1]} \\
E' & \xrightarrow{f'} & i^*X & \longrightarrow & D' & \longrightarrow & E'[1],
\end{array}$$

whose bottom row is the distinguished triangle on f'. The left square commutes because $i^*q_1 = f' \circ \theta$. By TR3, there exists $\mu\colon i^*N \to D'$ making the diagram commute. Since θ and the identity are isomorphisms, so is μ. By definition, aisles are strictly full, and so $i^*N \in \mathcal{D}_Z^{\le -1}$.

Next we treat the case on $\mathcal{D}_U$. By Lemma 4.2, $j^*N \in \mathcal{D}_U^{\leq K}$, and so $(j^*N)[K] \in \mathcal{D}_U^{\leq 0}$. Apply Lemma 3.4 to the weak approximation data $(G_U, \mathcal{D}_U^{\leq 0}, A)$ of Setup 4.4(2) and the object $(j^*N)[K]$, with $K+1$ in place of m. This gives a distinguished triangle

$$V' \to (j^*N)[K] \to R' \to V'[1]$$

where $V' \in \overline{\langle G_U \rangle}^{[-K-A,A]}$ and $R' \in \mathcal{D}_U^{\leq -K-1}$. Define $V := V'[-K]$ and $R := R'[-K]$. Shifting by $[-K]$ yields a distinguished triangle

$$V \xrightarrow{c} j^*N \to R \to V[1].$$

By Lemma 3.1(2),

$$V \in \overline{\langle G_U \rangle}^{[-A,A+K]} = \overline{\langle G_U \rangle}^{[-A,2A+1]}.$$

Moreover, $R \in \mathcal{D}_U^{\leq -K-1}[-K] = \mathcal{D}_U^{\leq -1}$.

Consider the composite

$$j_!V \xrightarrow{j_!(c)} j_!j^*N \xrightarrow{\epsilon_N} N, \qquad \psi : j_!V \to N.$$

Complete to a distinguished triangle

$$j_!V \xrightarrow{\psi} N \xrightarrow{\rho} D'' \to j_!V[1].$$

We claim that $D'' \in \mathcal{D}^{\leq -1}$. Since $j_!$ is exact, applying it to the distinguished triangle on c gives a distinguished triangle

$$j_!V \xrightarrow{j_!(c)} j_!j^*N \to j_!R \to j_!V[1].$$

Hence, $\operatorname{cone}(j_!(c)) \cong j_!R$. By (4.1), $\operatorname{cone}(\epsilon_N) \cong i_*i^*N$. Applying Remark 3.2(1) to the composite above gives a distinguished triangle

$$j_!R \to D'' \to i_*i^*N \to j_!R[1].$$

By Lemma 4.2, $R \in \mathcal{D}_U^{\leq -1}$ implies $j_!R \in \mathcal{D}^{\leq -1}$. Similarly, $i^*N \in \mathcal{D}_Z^{\leq -1}$ implies $i_*i^*N \in \mathcal{D}^{\leq -1}$. As aisles are closed under extensions, $D'' \in j_!R \star i_*i^*N \subseteq \mathcal{D}^{\leq -1}$.

We now assemble the two stages. Consider the composite

$$X \xrightarrow{r} N \xrightarrow{\rho} D'', \qquad \rho \circ r : X \to D''.$$

Complete to a distinguished triangle

$$E \to X \xrightarrow{\rho \circ r} D'' \to E[1].$$

The cocones of r, ρ, and $\rho \circ r$ are respectively E_1, $j_!V$, and E. Hence, Remark 3.2(2) gives a distinguished triangle $E_1 \to E \to j_!V \to E_1[1]$. Note that $E \in E_1 \star j_!V$.

Lastly we bound the range of E. Since $E_1 \in \mathcal{P}$ and $B = 3A$, $E_1 \in \overline{\langle G \rangle}^{[-3A,3A]}$. Note that $j_!$ is exact and preserves small coproducts. By Setup 4.4, $j_!G_U \in \overline{\langle G \rangle}^{[0,0]}$. By Lemma 3.1(3), $j_!V \in \overline{\langle G \rangle}^{[-A,2A+1]}$. As $2A+1 \leq 3A$, Lemma 3.1(2) gives $j_!V \in \overline{\langle G \rangle}^{[-3A,3A]}$. By Lemma 3.1(1), $\overline{\langle G \rangle}^{[-3A,3A]}$ is closed under extensions, and so $E \in \overline{\langle G \rangle}^{[-3A,3A]}$.

Thus, $E \to X \to D'' \to E[1]$ is a distinguished triangle with $E \in \overline{\langle G \rangle}^{[-3A,3A]}$ and $D'' \in \mathcal{D}^{\leq -1}$. As $X \in \mathcal{D}^{\leq 0}$ was arbitrary, $(G, \mathcal{D}^{\leq 0}, 3A)$ is weak approximation data for $\mathcal{D}$. Therefore, $\mathcal{D}$ is weakly approximable. □

Remark 5.1. The proof of Theorem 1.1 bootstraps that of [BNP23, Theorem 5.1]. The 'only if' direction is [BNP23, Remark 3]. For the 'if' direction, the two stages compare as follows.

- (On $\mathcal{D}_Z$) In [BNP23], the approximation $E' \to i^*\mathcal{X}$ has $E' \in \overline{\langle i^*G \rangle}_A^{[-A,A]}$ (see loc. cit. for notation). By [BNP23, Lemma 5.5], it can be lifted. In fact, loc. cit. proceeds by induction on the number of cones, which requires approximability. Here E' is only known to lie in $\overline{\langle i^*G \rangle}^{[-A,A]}$. Instead, it is lifted by Proposition 4.8.
- (On $\mathcal{D}_U$) The octahedra are those in the last paragraph of the proof of [BNP23, Theorem 5.1]. Here Lemma 3.4 is in place of the estimate for approximable categories used in loc. cit.

Remark 5.2. Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack. Pick a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Consider the following subcategories

- $D^{\leq 0}(\mathcal{X})$ consists of objects $E \in D(\mathcal{X})$ such that $\mathcal{H}^j(E) = 0$ for $j > 0$
- $D^{\geq 0}(\mathcal{X})$ consists of objects $E \in D(\mathcal{X})$ such that $\mathcal{H}^j(E) = 0$ for $j < 0$.

We call $(D^{\leq 0}(\mathcal{X}), D^{\geq 0}(\mathcal{X}))$ the **standard t-structure** on $D(\mathcal{X})$. The standard truncations of an object $E \in D_{\mathrm{qc}}(\mathcal{X})$ have quasi-coherent cohomology. Moreover, the support is defined via cohomology sheaves, so the truncations of an object supported on S are supported on S. See 'canonical truncations' in [Sta26, Tag 0118]. Hence, the standard t-structure induces a t-structure on $D_{\mathrm{qc},S}(\mathcal{X})$, which is denoted by $(D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X}), D^{\geq 0}_{\mathrm{qc},S}(\mathcal{X}))$. Write $D^{?}_{\mathrm{qc},S}(\mathcal{X})$ for $(D_{\mathrm{qc},S}(\mathcal{X}))^{?}$ in §2.1.3 with $? \in \{-, +, b\}$.

Definition 5.3. Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack. We say that $\mathcal{X}$ satisfies **W.A.** if $D_{\mathrm{qc},S}(\mathcal{X})$ is weakly approximable for every closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. We say $\mathcal{X}$ satisfies **standard W.A.** if for every closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement, there exists weak approximation data $(G, D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X}), A)$.

Remark 5.4. It is easy to see that standard W.A. implies W.A. in the sense of Definition 5.3.

Lemma 5.5. *Let $\mathcal{X}$ be a quasi-compact and quasi-separated algebraic stack. If G is a compact object of $D_{\mathrm{qc}}(\mathcal{X})$, then there exists an integer N such that $\mathrm{Hom}(G, G[n]) = 0$ for every $n > N$.*

Proof. By [HR17, Lemma 4.4(1)], G is perfect. Since $\mathcal{X}$ is quasi-compact, there exist $a \leq b$ such that $\mathcal{H}^j(G) \cong 0$ unless $j \in [a, b] \cap \mathbb{Z}$. Moreover, by [HR17, Lemma 4.5], there exists $r \geq 0$ such that $\mathrm{Hom}(G, \mathcal{F}[i]) = 0$ for every quasi-coherent $\mathcal{O}_{\mathcal{X}}$-module $\mathcal{F}$ and every $i > r$. For each $a \leq q \leq b$, there exists a distinguished triangle

$$\tau^{\leq q-1}G \to \tau^{\leq q}G \to \mathcal{H}^q(G)[-q] \to (\tau^{\leq q-1}G)[1].$$

Note that $\tau^{\leq a-1}G \cong 0$ and $\tau^{\leq b}G \cong G$. Fix $n > r + b$. For $a \leq q \leq b$, we have $n - q > r$. Since $\mathcal{H}^q(G)$ is quasi-coherent, it follows that

$$\begin{aligned}\mathrm{Hom}(G, \mathcal{H}^q(G)[-q+n]) &= \mathrm{Hom}(G, \mathcal{H}^q(G)[n-q]) \\ &= 0.\end{aligned}$$

We prove by induction on q that $\mathrm{Hom}(G, (\tau^{\leq q}G)[n]) = 0$ for every $a-1 \leq q \leq b$. The case for $q = a-1$ follows from $\tau^{\leq a-1}G \cong 0$. For $a \leq q \leq b$, applying $\mathrm{Hom}(G, -)$ to the distinguished triangle above yields an exact sequence

$$\mathrm{Hom}(G, (\tau^{\leq q-1}G)[n]) \to \mathrm{Hom}(G, (\tau^{\leq q}G)[n]) \to \mathrm{Hom}(G, \mathscr{H}^q(G)[n-q]).$$

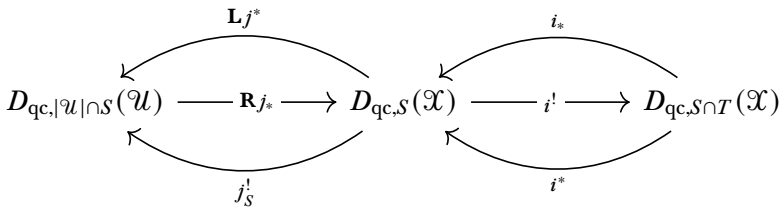
By the induction hypothesis, the first group vanishes. By discussion above, the last group vanishes. By exactness, the middle group vanishes. Taking $q = b$ and using $\tau^{\leq b}G \cong G$, it follows that $\mathrm{Hom}(G, G[n]) = 0$ for every $n > r + b$. This completes the proof. □

Lemma 5.6. *Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack. Choose $S \subseteq |\mathcal{X}|$ closed with quasi-compact complement. If G' is a compact generator of $D_{\mathrm{qc},S}(\mathcal{X})$, then $\mathrm{Hom}(G', G'[n]) = 0$ for $n \gg 0$.*

Proof. By [Lan26, Remark 3.2], G' is compact in $D_{\mathrm{qc}}(\mathcal{X})$. By Lemma 5.5, the claim follows. □

Proposition 5.7. *Consider an étale neighborhood $j\colon \mathcal{U} \to \mathcal{X}$ and $f\colon \mathcal{Y} \to \mathcal{X}$ of 1-Thomason algebraic stacks. If $\mathcal{U}$ and $\mathcal{Y}$ satisfy (resp. standard) W.A., then so does $\mathcal{X}$.*

Proof. Fix a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Set $T := |\mathcal{X}| \setminus |\mathcal{U}|$. By [Lan26, Proposition 3.1], there exists a recollement



where i_* is the natural inclusion. We apply Theorem 1.1 to this recollement. In the notation of Setup 4.1, the roles are as follows:

- $\mathscr{D}_Z = D_{\mathrm{qc},|\mathcal{U}|\cap S}(\mathcal{U})$, where $\mathbf{L}j^*$, $\mathbf{R}j_*$, $j_S^!$ (of the diagram above) are respectively identified with i^*, i_*, $i^!$
- $\mathscr{D} = D_{\mathrm{qc},S}(\mathcal{X})$
- $\mathscr{D}_U = D_{\mathrm{qc},S\cap T}(\mathcal{X})$, where i_*, $i^!$, i^* are respectively identified with $j_!$, j^*, j_* (in Setup 4.1).

First, $|\mathcal{U}| \setminus (|\mathcal{U}| \cap S) = |\mathcal{U}| \cap (|\mathcal{X}| \setminus S)$. Since $|\mathcal{U}| \cap (|\mathcal{X}| \setminus S)$ is the intersection of two quasi-compact opens of the quasi-separated topological space $|\mathcal{X}|$, it is quasi-compact. As $\mathcal{U}$ satisfies W.A., $\mathscr{D}_Z$ is weakly approximable.

Next, apply [HR23, Proposition 4.2] to $T \cap S$. This gives a triangulated equivalence

$$\mathbf{L}f^*\colon D_{\mathrm{qc},T\cap S}(\mathcal{X}) \to D_{\mathrm{qc},f^{-1}(T\cap S)}(\mathcal{Y}).$$

Note that $|\mathcal{Y}| \setminus f^{-1}(T \cap S) = f^{-1}(|\mathcal{U}| \cup (|\mathcal{X}| \setminus S))$. Since $\mathcal{Y}$ is quasi-compact and $\mathcal{X}$ is quasi-separated, f is quasi-compact [Sta26, Tag 075S]. As $|\mathcal{U}| \cup (|\mathcal{X}| \setminus S)$ is a quasi-compact open, $f^{-1}(|\mathcal{U}| \cup (|\mathcal{X}| \setminus S))$ is quasi-compact. Moreover, as $\mathcal{Y}$ satisfies W.A., $D_{\mathrm{qc},f^{-1}(T\cap S)}(\mathcal{Y})$ is weakly approximable. By [CNS25, Remark 3.4(i)], weak approximability is invariant under triangulated equivalence. Thus, $\mathscr{D}_U$ is weakly approximable.

Observe that $|\mathcal{X}| \setminus (S \cap T) = (|\mathcal{X}| \setminus S) \cup |\mathcal{U}|$, which is quasi-compact. Moreover, by the 1-Thomason condition, the three triangulated categories in the recollement are singly

compactly generated. If G' is a compact generator of $D_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$, then Lemma 5.6 says that $\mathrm{Hom}(G', G'[n]) = 0$ for $n \gg 0$. Consequently, Theorem 1.1 asserts that $D_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$ is weakly approximable.

Suppose that $\mathcal{U}$ and $\mathcal{Y}$ satisfy standard W.A. By [Nee26, Facts 1.26(i)], the standard t-structures on the two outside terms lie in their respective preferred equivalence classes. For the term supported on $T \cap S$, use the t-exact equivalence induced by $\mathbf{L}f^*$. Denote by $\mathcal{D}_{\mathrm{gl}}^{\leq 0}$ the aisle obtained by gluing these two standard t-structures.

In the proof of Theorem 1.1, the t-structure used on $D_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$ is glued from t-structures in the preferred equivalence classes of the two outside terms. See Setup 4.4. Replacing either outside t-structure by an equivalent one changes the glued t-structure only up to equivalence. Indeed, this follows from the description of its aisle in Lemma 4.2, with the maximum of the two bounds. Hence, $\mathcal{D}_{\mathrm{gl}}^{\leq 0}$ is in the preferred equivalence class of $D_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$.

We compare this glued t-structure with the standard one. By Lemma 5.11, there exists $r \geq 0$ such that $\mathbf{R}j_* D^{\leq 0}_{\mathrm{qc},|\mathcal{U}|\cap\mathcal{S}}(\mathcal{U}) \subseteq D^{\leq r}_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$. If $E \in \mathcal{D}_{\mathrm{gl}}^{\leq 0}$, the recollement triangle

$$i_* i^! E \to E \to \mathbf{R}j_* \mathbf{L}j^* E \to (i_* i^! E)[1]$$

implies $E \in D^{\leq r}_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$. Conversely, if $E \in D^{\leq 0}_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$, the same distinguished triangle implies $i^! E \in D^{\leq r+1}_{\mathrm{qc},\mathcal{S}\cap\mathcal{T}}(\mathcal{X})$. As $\mathbf{L}j^* E \in D^{\leq 0}_{\mathrm{qc},|\mathcal{U}|\cap\mathcal{S}}(\mathcal{U})$, it follows that

$$D^{\leq 0}_{\mathrm{qc},\mathcal{S}}(\mathcal{X}) \subseteq \mathcal{D}_{\mathrm{gl}}^{\leq r+1} \subseteq D^{\leq 2r+1}_{\mathrm{qc},\mathcal{S}}(\mathcal{X}).$$

Hence, the glued and standard t-structures are equivalent. Since the standard t-structure belongs to the preferred class, [BNP23, Lemma 3.7] gives the desired weak approximation data. Therefore, $\mathcal{X}$ satisfies standard W.A. Since $\mathcal{S}$ was arbitrary, the first assertion gives W.A. and the other argument gives standard W.A. □

Lemma 5.8. *Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack with quasi-finite separated diagonal that satisfies standard W.A. Fix a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Set $\mathcal{T} := D_{\mathrm{qc},\mathcal{S}}(\mathcal{X})$. Then $\mathcal{T}_c^-$ consists of the objects of $\mathcal{T}$ that are pseudocoherent in $D(\mathcal{X})$ and supported on S. If $\mathcal{X}$ is Noetherian, $\mathcal{T}_c^? = D^?_{\mathrm{coh},\mathcal{S}}(\mathcal{X})$ for $? \in \{-, b\}$.*

Proof. Since the standard t-structure appears in weak approximation data, [Nee26, Facts 1.26(i)] implies it lies in the preferred equivalence class. Hence, we compute $\mathcal{T}_c^-$ and $\mathcal{T}_c^b$ with respect to it.

Pick $E \in \mathcal{T}_c^-$ and $n > 0$. Choose a distinguished triangle

$$D \to E \to F \to D[1]$$

with $D \in \mathcal{T}^c$ and $F \in \mathcal{T}^{\leq -n}$. By [HR17, Lemma 4.4], D is perfect, and hence, pseudocoherent [Sta26, Tag 08G8]. Since $\mathcal{H}^j(F) = 0$ for $j \geq 1 - n$, F is $(1-n)$-pseudocoherent. Hence, E is $(1-n)$-pseudocoherent [Sta26, Tag 08FV]. As n was arbitrary, E is pseudocoherent.

Conversely, let $E \in \mathcal{T}$ be pseudocoherent. Fix $n > 0$. The open immersion with image $|\mathcal{X}| \setminus S$ is quasi-compact [Sta26, Tag 075S]. By [HLLP25, Theorem A], there exist a compact object P supported on S and $P \to E$ inducing isomorphisms on $\mathcal{H}^i$ for $i > 1 - n$ and a surjection on $\mathcal{H}^{1-n}$. Note that the cone of $P \to E$ lies in $\mathcal{T}^{\leq -n}$. Clearly, $P \in \mathcal{T}^c$. Therefore, $E \in \mathcal{T}_c^-$, which gives the first claim.

We show the second claim. By [Sta26, Tag 08IK], the pseudocoherent objects of $D_{\mathrm{qc}}(\mathcal{X})$ are exactly those of $D^-_{\mathrm{coh}}(\mathcal{X})$. Indeed, this can be checked by pulling back on a smooth

presentation. Hence, $\mathcal{T}_c^- = D^-_{\mathrm{coh}}(\mathcal{X}) \cap \mathcal{T} = D^-_{\mathrm{coh},S}(\mathcal{X})$, and $\mathcal{T}_c^b = \mathcal{T}^b \cap \mathcal{T}_c^- = D^b_{\mathrm{coh},S}(\mathcal{X})$. This completes the proof. □

Lemma 5.9. *Let $f\colon \mathcal{Y} \to \mathcal{X}$ be a finite, faithfully flat morphism of finite presentation between quasi-compact quasi-separated algebraic stacks. Choose a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Set $T := f^{-1}(S)$. Then there is an adjunction*

$$\mathbf{R}f_*\colon D_{\mathrm{qc},T}(\mathcal{Y}) \leftrightarrows D_{\mathrm{qc},S}(\mathcal{X})\colon f^\times$$

satisfying the following properties:

1. *Both functors preserve small coproducts and are t-exact for the standard t-structures.*
2. *The functor $f^\times$ is conservative.*
3. *For every $E \in D_{\mathrm{qc},S}(\mathcal{X})$, the counit $\mathbf{R}f_*f^\times E \to E$ induces surjections on all cohomology sheaves.*
4. *If P is a compact generator of $D_{\mathrm{qc},T}(\mathcal{Y})$, then $\mathbf{R}f_*P$ is a compact generator of $D_{\mathrm{qc},S}(\mathcal{X})$.*

Proof. First we recall the relevant facts. Since f is finite, it is representable and affine, and hence concentrated [HR17, Lemma 2.5(3)]. By [HR17, Theorem 2.6(3)(4)], $\mathbf{R}f_*\colon D_{\mathrm{qc}}(\mathcal{Y}) \to D_{\mathrm{qc}}(\mathcal{X})$ preserves small coproducts and is compatible with flat base change on $\mathcal{X}$. By [HR17, Corollary 4.15], $\mathbf{R}f_*$ admits a right adjoint $f^\times$ which preserves small coproducts, is conservative, and is compatible with base change on $\mathcal{X}$. Here the base change isomorphism for $f^\times$ is obtained by adjunction from the one for $\mathbf{R}f_*$ (see the proof of [HR17, Theorem 4.14(4)]).

Next we restrict to the supported categories. Let $j\colon \mathcal{U} \to \mathcal{X}$ be the quasi-compact open immersion with $|\mathcal{U}| = |\mathcal{X}| \setminus S$. Form the fibered square

$$\begin{array}{ccc} \mathcal{V} & \xrightarrow{j'} & \mathcal{Y} \\ {\scriptstyle f_{\mathcal{U}}}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{U} & \xrightarrow[j]{} & \mathcal{X}. \end{array}$$

Here $|\mathcal{V}| = |\mathcal{Y}| \setminus T$. Since j is flat, there are natural isomorphisms $\mathbf{L}j^*\mathbf{R}f_* \cong \mathbf{R}(f_{\mathcal{U}})_*\mathbf{L}(j')^*$ and $\mathbf{L}(j')^*f^\times \cong (f_{\mathcal{U}})^\times \mathbf{L}j^*$. Consequently, $\mathbf{R}f_*$ sends objects supported on T to objects supported on S, and $f^\times$ sends objects supported on S to objects supported on T. Therefore, the adjunction restricts to the asserted supported categories. Note that $D_{\mathrm{qc},T}(\mathcal{Y})$ and $D_{\mathrm{qc},S}(\mathcal{X})$ are the kernels of the coproduct preserving functors $\mathbf{L}(j')^*$ and $\mathbf{L}j^*$. Hence, they are respectively closed under small coproducts in $D_{\mathrm{qc}}(\mathcal{Y})$ and $D_{\mathrm{qc}}(\mathcal{X})$. It follows that both restricted functors preserve small coproducts, and that the restriction of $f^\times$ is conservative. This proves (2), as well as the claim on coproducts in (1).

Now we work smooth locally on $\mathcal{X}$. Choose a smooth surjective morphism $s\colon \mathrm{Spec}(R) \to \mathcal{X}$. Since f is finite, $\mathrm{Spec}(R) \times_{\mathcal{X}} \mathcal{Y}$ is affine, say $\mathrm{Spec}(B)$ for a finite R-algebra B. Consider the fibered square

$$\begin{array}{ccc} \mathrm{Spec}(B) & \xrightarrow{t} & \mathcal{Y} \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(R) & \xrightarrow[s]{} & \mathcal{X}. \end{array}$$

The morphism g is faithfully flat and of finite presentation. Hence, by [Sta26, Tag 02KB], B is a finite projective R-module. This ensures that B is faithfully flat over R. Since s is flat, there are natural isomorphisms $\mathbf{L}s^*\mathbf{R}f_* \cong \mathbf{R}g_*\mathbf{L}t^*$ and $\mathbf{L}t^*f^\times \cong g^\times \mathbf{L}s^*$.

Under the equivalences $D_{\mathrm{qc}}(\mathrm{Spec}(R)) \cong D(R)$ and $D_{\mathrm{qc}}(\mathrm{Spec}(B)) \cong D(B)$, $\mathbf{R}g_*$ is restriction of scalars along $R \to B$. Indeed, g is affine, so [Sta26, Tag 0G9R] identifies derived pushforward with ordinary pushforward on complexes of quasi-coherent modules. In particular, $\mathbf{R}g_*$ is t-exact. By [Sta26, Tag 0A70], the right adjoint of restriction of scalars is $\mathbf{R}\operatorname{Hom}_R(B,-)$. As B is finite projective over R, [Sta26, Tag 05CE] implies that $\operatorname{Hom}_R(B,-)$ is exact. Hence, by [Sta26, Tag 0A71], $g^\times$ is identified with

$$\mathbf{R}\operatorname{Hom}_R(B,-) \cong \operatorname{Hom}_R(B,R) \otimes_R -,$$

where the last isomorphism follows from finite projectivity. As $\operatorname{Hom}_R(B,R)$ is a finite projective R-module, $g^\times$ is t-exact.

Since s and t are smooth and surjective, $\mathbf{L}s^*$ and $\mathbf{L}t^*$ are t-exact and conservative. In particular, an object $E \in D_{\mathrm{qc}}(\mathcal{X})$ lies in $D_{\mathrm{qc}}^{\leq 0}(\mathcal{X})$ (resp. $D_{\mathrm{qc}}^{\geq 0}(\mathcal{X})$) if, and only if, $\mathbf{L}s^*E$ lies in $D_{\mathrm{qc}}^{\leq 0}(\mathrm{Spec}(R))$ (resp. $D_{\mathrm{qc}}^{\geq 0}(\mathrm{Spec}(R))$), and similarly for t. The base change isomorphisms above now imply that $\mathbf{R}f_*$ and $f^\times$ are t-exact. As the standard t-structures on the supported categories are induced from those on D_{qc}, the restricted functors are t-exact. This completes the proof of (1).

Next we prove (3). Fix $E \in D_{\mathrm{qc}}(\mathcal{X})$. Denote by $\varepsilon_E \colon \mathbf{R}f_*f^\times E \to E$ the counit. Since the base change isomorphism for $f^\times$ is obtained by adjunction from the one for $\mathbf{R}f_*$, the triangle identities show that, up to these isomorphisms on its source, $\mathbf{L}s^*\varepsilon_E$ is the counit $\mathbf{R}g_*g^\times \mathbf{L}s^*E \to \mathbf{L}s^*E$ of $\mathbf{R}g_* \dashv g^\times$. Under the equivalence with $D(R)$, the latter is the morphism

$$\operatorname{Hom}_R(B,R) \otimes_R \mathbf{L}s^*E \to \mathbf{L}s^*E$$

induced by evaluation at $1 \in B$.

We claim that evaluation at 1 induces a surjection $\operatorname{Hom}_R(B,R) \to R$. Indeed, let $\mathfrak{m} \subseteq R$ be a maximal ideal. Since B is faithfully flat over R, $B/\mathfrak{m}B$ is nonzero, and so the image of $1 \in B$ in $B/\mathfrak{m}B$ is nonzero. As $B_{\mathfrak{m}}$ is finite free over the local ring $R_{\mathfrak{m}}$, Nakayama's lemma implies that 1 extends to a basis of $B_{\mathfrak{m}}$. Hence, there exists an $R_{\mathfrak{m}}$-linear map $B_{\mathfrak{m}} \to R_{\mathfrak{m}}$ sending 1 to 1. As B is a finitely presented R-module, $\operatorname{Hom}_R(B,R)_{\mathfrak{m}} \cong \operatorname{Hom}_{R_{\mathfrak{m}}}(B_{\mathfrak{m}}, R_{\mathfrak{m}})$. Thus, evaluation at 1 is surjective after localizing at every maximal ideal of R, and hence it is surjective.

Since $\operatorname{Hom}_R(B,R)$ is a projective R-module, $\operatorname{Hom}_R(B,R) \otimes_R -$ is exact. Hence, for every $n \in \mathbb{Z}$, $\mathcal{H}^n(\mathbf{L}s^*\varepsilon_E)$ is identified with the tensor product of the surjection above with $\mathcal{H}^n(\mathbf{L}s^*E)$. By right exactness of the tensor product, $\mathcal{H}^n(\mathbf{L}s^*\varepsilon_E)$ is surjective. As s is flat, $s^*\mathcal{H}^n(\varepsilon_E) \cong \mathcal{H}^n(\mathbf{L}s^*\varepsilon_E)$, and so the pullback along s of the cokernel of $\mathcal{H}^n(\varepsilon_E)$ vanishes. Since s is faithfully flat, $\mathcal{H}^n(\varepsilon_E)$ is surjective. This proves (3).

Lastly, let P be a compact generator of $D_{\mathrm{qc},T}(\mathcal{Y})$. Since the restriction of $f^\times$ preserves small coproducts, adjunction implies that $\mathbf{R}f_*P$ is compact in $D_{\mathrm{qc},S}(\mathcal{X})$. If $E \in D_{\mathrm{qc},S}(\mathcal{X})$ satisfies $\operatorname{Hom}(\mathbf{R}f_*P, E[n]) = 0$ for every $n \in \mathbb{Z}$, then adjunction gives $\operatorname{Hom}(P, f^\times E[n]) = 0$ for every $n \in \mathbb{Z}$. Since P is a compact generator, $f^\times E \cong 0$. By conservativity, $E \cong 0$. Hence, $\mathbf{R}f_*P$ is a compact generator, which proves (4). □

Lemma 5.10. *Let $f : \mathcal{Y} \to \mathcal{X}$ be a finite, faithfully flat morphism of finite presentation between quasi-compact quasi-separated algebraic stacks. If $\mathcal{Y}$ satisfies standard W.A., then so does $\mathcal{X}$.*

Proof. Fix a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Set $T := f^{-1}(S)$. Since f is quasi-compact, $|\mathcal{Y}| \setminus T = f^{-1}(|\mathcal{X}| \setminus S)$ is quasi-compact. As $\mathcal{Y}$ satisfies standard W.A., there exists weak approximation data $(P, D^{\leq 0}_{\mathrm{qc},T}(\mathcal{Y}), A)$. By Lemma 5.9(4), $G := \mathbf{R}f_*P$ is a compact generator of $D_{\mathrm{qc},S}(\mathcal{X})$. We show that $(G, D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X}), A)$ is weak approximation data for $D_{\mathrm{qc},S}(\mathcal{X})$.

We start with Definition 2.1(1). Since $P \in D^{\leq A}_{\mathrm{qc},T}(\mathcal{Y})$ and $\mathbf{R}f_*$ is t-exact, $G \in D^{\leq A}_{\mathrm{qc},S}(\mathcal{X})$. Let $F \in D^{\leq -A}_{\mathrm{qc},S}(\mathcal{X})$. Since $f^\times$ is t-exact, $f^\times F \in D^{\leq -A}_{\mathrm{qc},T}(\mathcal{Y})$. Hence, by adjunction,

$$\mathrm{Hom}(G, F) \cong \mathrm{Hom}(P, f^\times F) = 0.$$

Now we verify Definition 2.1(2). Let $E \in D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X})$. As $f^\times E \in D^{\leq 0}_{\mathrm{qc},T}(\mathcal{Y})$, there exists a distinguished triangle

$$Q \xrightarrow{a} f^\times E \to D \to Q[1]$$

where $Q \in \overline{\langle P \rangle}^{[-A,A]}$ and $D \in D^{\leq -1}_{\mathrm{qc},T}(\mathcal{Y})$. Since $\mathbf{R}f_*$ is exact, preserves small coproducts, and satisfies $\mathbf{R}f_*P = G$, Lemma 3.1(3) implies $\mathbf{R}f_*Q \in \overline{\langle G \rangle}^{[-A,A]}$. Denote by $\varepsilon_E \colon \mathbf{R}f_*f^\times E \to E$ the counit, and consider the composite

$$\mathbf{R}f_*Q \xrightarrow{\mathbf{R}f_*a} \mathbf{R}f_*f^\times E \xrightarrow{\varepsilon_E} E.$$

It suffices to show that its cone belongs to $D^{\leq -1}_{\mathrm{qc},S}(\mathcal{X})$. By Remark 3.2(1), there is a distinguished triangle

$$\mathrm{cone}(\mathbf{R}f_*a) \to \mathrm{cone}(\varepsilon_E \circ \mathbf{R}f_*a) \to \mathrm{cone}(\varepsilon_E) \to \mathrm{cone}(\mathbf{R}f_*a)[1].$$

As aisles are closed under extensions, it suffices to show that $\mathrm{cone}(\mathbf{R}f_*a)$ and $\mathrm{cone}(\varepsilon_E)$ belong to $D^{\leq -1}_{\mathrm{qc},S}(\mathcal{X})$. Since $\mathbf{R}f_*$ is exact, $\mathrm{cone}(\mathbf{R}f_*a) \cong \mathbf{R}f_*D$, which belongs to $D^{\leq -1}_{\mathrm{qc},S}(\mathcal{X})$ by t-exactness of $\mathbf{R}f_*$. By t-exactness of $\mathbf{R}f_*$ and $f^\times$, $\mathbf{R}f_*f^\times E \in D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X})$. Moreover, $\mathcal{H}^0(\varepsilon_E)$ is surjective by Lemma 5.9(3). Hence, the long exact sequence of cohomology sheaves shows that $\mathrm{cone}(\varepsilon_E) \in D^{\leq -1}_{\mathrm{qc},S}(\mathcal{X})$. This shows Definition 2.1(2) for $(G, D^{\leq 0}_{\mathrm{qc},S}(\mathcal{X}), A)$. As S was arbitrary, $\mathcal{X}$ satisfies standard W.A. □

Lemma 5.11. *Let $j \colon \mathcal{U} \to \mathcal{X}$ be a quasi-compact open immersion between quasi-compact quasi-separated algebraic stacks. Then $\mathbf{R}j_* \colon D_{\mathrm{qc}}(\mathcal{U}) \to D_{\mathrm{qc}}(\mathcal{X})$ preserves small coproducts. There is an integer $r \geq 0$ such that*

$$\mathbf{R}j_*D^{\leq 0}_{\mathrm{qc}}(\mathcal{U}) \subseteq D^{\leq r}_{\mathrm{qc}}(\mathcal{X}).$$

For any quasi-compact open subset $V \subseteq |\mathcal{U}|$, the same statements hold for the restricted functor $\mathbf{R}j_* \colon D_{\mathrm{qc},|\mathcal{U}|\setminus V}(\mathcal{U}) \to D_{\mathrm{qc},|\mathcal{X}|\setminus V}(\mathcal{X})$.

Proof. By [HR17, Theorem 2.6], $\mathbf{R}j_*$ preserves small coproducts. Let $s \colon V \to \mathcal{X}$ be a smooth surjective morphism by an affine scheme. Consider the fibered square

$$\begin{array}{ccc} \mathcal{U} \times_{\mathcal{X}} V & \xrightarrow{j'} & V \\ {\scriptstyle s'}\downarrow & & \downarrow{\scriptstyle s} \\ \mathcal{U} & \xrightarrow[j]{} & \mathcal{X}. \end{array}$$

As j is representable by schemes, $\mathcal{U} \times_{\mathcal{X}} V$ is a scheme. Base change implies j' is an open immersion, and that s' is a smooth surjection. By [Lip09, Proposition 3.9.2], there is an integer $r \geq 0$ such that

$$\mathbf{R}j_* D_{\mathrm{qc}}^{\leq 0}(\mathcal{U} \times_{\mathcal{X}} V) \subseteq D_{\mathrm{qc}}^{\leq r}(V).$$

Pick $E \in D_{\mathrm{qc}}^{\leq 0}(\mathcal{U})$. By [HR17, Theorem 2.6(4)], $\mathbf{L}s^* \mathbf{R}j_* E \cong \mathbf{R}j'_* \mathbf{L}(s')^* E$. It follows that $\mathbf{R}j'_* \mathbf{L}(s')^* E \in D_{\mathrm{qc}}^{\leq r}(V)$ implies $\mathbf{R}j_* E \in D_{\mathrm{qc}}^{\leq r}(\mathcal{X})$. This proves the first claim, whereas the last follows from supports. □

Lemma 5.12. *Let $j\colon \mathcal{U} \to \mathcal{X}$ be a quasi-compact open immersion of quasi-compact quasi-separated algebraic stacks. If $\mathcal{X}$ satisfies standard W.A., then so does $\mathcal{U}$.*

Proof. Choose a closed subset $S \subseteq |\mathcal{U}|$ with quasi-compact complement $V := |\mathcal{U}| \setminus S$. Since V is a quasi-compact open subset of $|\mathcal{X}|$, the closed subset $T := |\mathcal{X}| \setminus V$ has quasi-compact complement. Note that $j^{-1}(T) = S$. Fix weak approximation data $(G, D_{\mathrm{qc},T}^{\leq 0}(\mathcal{X}), A)$. Note that $\mathbf{R}j_*\colon D_{\mathrm{qc},S}(\mathcal{U}) \to D_{\mathrm{qc},T}(\mathcal{X})$ is conservative and preserves small coproducts. Hence, by [Lan26, Corollary 3.6], $\mathbf{L}j^* G$ is a compact generator of $D_{\mathrm{qc},S}(\mathcal{U})$.

By Lemma 5.11, there exists $r \geq 0$ such that $\mathbf{R}j_* D_{\mathrm{qc},S}^{\leq 0}(\mathcal{U}) \subseteq D_{\mathrm{qc},T}^{\leq r}(\mathcal{X})$. For $E \in D_{\mathrm{qc},S}^{\leq -A-r}(\mathcal{U})$, adjunction gives

$$\mathrm{Hom}(\mathbf{L}j^* G, E) \cong \mathrm{Hom}(G, \mathbf{R}j_* E) = 0.$$

Moreover, $\mathbf{L}j^* G \in D_{\mathrm{qc},S}^{\leq A}(\mathcal{U})$.

Now let $E \in D_{\mathrm{qc},S}^{\leq 0}(\mathcal{U})$. Then $(\mathbf{R}j_* E)[r] \in D_{\mathrm{qc},T}^{\leq 0}(\mathcal{X})$. By Lemma 3.4, applied with $m = r + 1$, there exists a distinguished triangle

$$P \to (\mathbf{R}j_* E)[r] \to D \to P[1]$$

where $P \in \overline{\langle G \rangle}^{[-A-r, A]}$ and $D \in D_{\mathrm{qc},T}^{\leq -r-1}(\mathcal{X})$. After shifting by $[-r]$ and applying the exact functor $\mathbf{L}j^*$, we obtain

$$\mathbf{L}j^*(P[-r]) \to E \to \mathbf{L}j^*(D[-r]) \to (\mathbf{L}j^*(P[-r]))[1].$$

Indeed, composite of $\mathbf{L}j^*$ with $\mathbf{R}j_*$ is naturally isomorphic to the identity. Here the first term belongs to $\overline{\langle \mathbf{L}j^* G \rangle}^{[-A-r, A+r]}$ and the third belongs to $D_{\mathrm{qc},S}^{\leq -1}(\mathcal{U})$. Increasing the uniform constant, if necessary, yields standard weak approximation data for $D_{\mathrm{qc},S}(\mathcal{U})$. As S was arbitrary, $\mathcal{U}$ satisfies standard W.A. □

Proof of Corollary 1.2. We establish standard W.A., which is stronger than the first claim. Fix $\mathcal{X}$ as in the statement. Let $\mathscr{C}$ be the strictly full 2-subcategory of algebraic stacks over $\mathcal{X}$ whose structure morphisms are representable, separated, finitely presented, quasi-finite, and flat. Note that every object of $\mathscr{C}$ is quasi-compact quasi-separated and has quasi-finite separated diagonal. Hence, by [HR17, Theorem A], every object of $\mathscr{C}$ satisfies the 1-Thomason condition. Set $\mathscr{C} \subseteq \mathscr{C}$ to be the strictly full 2-subcategory consisting of stacks satisfying standard W.A.

We verify the three hypotheses of the induction principle [HR18, Theorem E]. First, by Lemma 5.12, standard W.A. passes to quasi-compact open substacks. Second, suppose $f\colon V \to \mathcal{W}$ is finite, faithfully flat and of finite presentation with V affine. By [Nee24, Theorem 3.2(iii,iv)], V satisfies standard W.A. Hence, by Lemma 5.10, $\mathcal{W}$ satisfies standard W.A. Third, given an étale neighborhood $\mathcal{U} \to \mathcal{W} \leftarrow \mathcal{Y}$ in $\mathscr{C}$, standard W.A. for $\mathcal{U}$ and $\mathcal{Y}$

implies standard W.A. for $\mathscr{W}$ by Proposition 5.7. Thus, [HR18, Theorem E] gives $\mathscr{C} = \mathscr{E}$, and so $\mathcal{X}$ satisfies standard W.A. Consequently, the first claim follows, and the case $S = |\mathcal{X}|$ gives the second.

We prove the last claim. Here $\mathcal{X}$ is Noetherian. Fix a closed subset $S \subseteq |\mathcal{X}|$ with quasi-compact complement. Set $\mathcal{T} := D_{\mathrm{qc},S}(\mathcal{X})$. By Lemma 5.8, $\mathcal{T}_c^- = D^-_{\mathrm{coh},S}(\mathcal{X})$ and $\mathcal{T}_c^b = D^b_{\mathrm{coh},S}(\mathcal{X})$. The standard t-structure restricts to $D^-_{\mathrm{coh},S}(\mathcal{X})$. Thus, we finish by [CHNS24, Example 10.1.2] and the first claim. Indeed, by loc. cit., a weakly approximable category is coherent if some t-structure in the preferred class restricts to $\mathcal{T}_c^-$. □

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